Beyond the Training Distribution: A Survey of Reasoning Generalization in Large Language Models

Anonymous ACL submission

Abstract

Large language models (LLMs) have achieved strong performance on mathematical, logical, and scientific reasoning tasks. However, this progress is commonly measured on benchmarks that remain close to the distributions encountered during training. Small changes in notation or problem structure, as well as deeper reasoning or new domains, can lead to substantial performance degradation, raising a central question: *when and how can LLMs generalize their reasoning beyond their training distributions?* This paper presents a comprehensive survey of reasoning generalization in LLMs. We organize existing work into four complementary pillars: (1) training methods that encourage transferable reasoning, (2) inference strategies that adapt computation at test time, (3) architectural designs that support reusable operations, and (4) analyses that characterize when and why generalization succeeds or fails. Building on this taxonomy, we compare the assumptions, evaluation settings, and generalization claims of representative approaches. We further summarize the main empirical findings and theoretical results, and discuss the evaluation practices and open challenges that shape this emerging research direction.

1 Introduction

Recent advances in prompting, post-training, and test-time computation have improved the reasoning capabilities of large language models (LLMs). Models can now solve difficult mathematical problems, construct multi-step proofs, and use external tools. These advances raise a central question: when and how can LLMs generalize their reasoning beyond their training distributions?

Strong benchmark performance does not resolve this question. Accuracy can vary across equivalent numerical versions of a problem and decline when irrelevant clauses are added (Mirzadeh et al., 2025). Models also struggle with deeper or wider proof trees (Opedal et al., 2025) and unfamiliar combinations of scientific domains (Gong et al., 2026). Similar failures occur under symbol remapping, longer inputs, new task compositions, and changed tool interfaces. These findings suggest that success on held-out examples alone does not establish that a model has learned a reusable reasoning procedure.

Assessing reasoning generalization requires a clear account of what is "unseen." A task absent from supervised fine-tuning may still resemble pretraining data, while extra samples or tool calls may extend system performance without improving the base model. Studies also differ in scope, ranging from evaluations of frontier LLMs to experiments on small transformers and formal results under restricted assumptions. Comparisons therefore need to state what the model has encountered, what changes at test time, and which components remain fixed.

Efforts to improve reasoning generalization mainly span training, inference, and architecture. Training methods seek transferable procedures through data, objectives, and curricula. Inference methods use structured reasoning, additional computation, verification, and external memory to apply a fixed model to unfamiliar tasks. Architectural studies investigate recurrent depth, information routing, and recurrent memory. Analytical studies examine the resulting behavior, internal mechanisms, and theoretical limits. Existing surveys cover broad NLP generalization (Hupkes et al., 2023), out-of-distribution learning (Liu et al., 2021), inductive reasoning (Chen et al., 2025b), or generalization in language-model agents (Zhang et al., 2025b). Appendix B.2 compares our scope with related surveys of latent reasoning, logical reasoning, and planning. Our focus is the transfer of reasoning procedures across explicit distribution shifts.

This survey brings these research directions together. As shown in Figure 1, we organize the

literature into four pillars: *Training for Generalization* (§3), *Inference for Generalization* (§4), *Architecture for Generalization* (§5), and *Analysis of Generalization* (§6). Together, these pillars organize methods that support reasoning generalization and evidence about the conditions under which it succeeds or fails.

Our key contributions are threefold: (1) **Systematic Taxonomy.** We present a unified taxonomy that organizes studies of reasoning generalization into four pillars (Figure 1), clarifying their scope and objectives. (2) **In-depth Discussion.** Building on this taxonomy, we analyze representative works, comparing training strategies, inference procedures, architectural designs, and behavioral, mechanistic, and theoretical evidence to examine when and how reasoning generalizes. (3) **Challenges and Future Directions.** We identify key challenges in transfer, length and depth extrapolation, and interpretability, and outline promising research directions in transferable training, stable recurrent computation, and verification (§7).

2 Preliminaries

Definitions. Let x denote a reasoning problem, y its answer, and a the inference procedure applied by a model with parameters θ. The resulting system is written as $f_{\theta,a}(x)$, with expected risk $R_P(\theta, a) = \mathbb{E}_{(x,y)\sim P}[\ell(f_{\theta,a}(x), y)]$ on a distribution P. Evaluation is in-distribution (ID) when training and test examples follow the same relevant data-generating process, and out-of-distribution (OOD) when a specified property of that process changes (Liu et al., 2021). When the two losses are comparable, the transfer gap is $R_{\mathrm{OOD}}(\theta, a) - R_{\mathrm{ID}}(\theta, a)$. We use *reasoning generalization* to describe the ability to preserve a reasoning procedure under such a change. This includes compositional generalization to new combinations of familiar primitives (Lake and Baroni, 2018) and length generalization to longer or deeper instances (Anil et al., 2022).

Overview. Following Figure 1, *Training for Generalization* (§3) reviews methods that update model parameters, *Inference for Generalization* (§4) covers methods that operate on a fixed model, and *Architecture for Generalization* (§5) studies designs that support reusable computation. Beyond these methods, *Analysis of Generalization* (§6) examines what transfers and why. Methods spanning several pillars are discussed under their primary source of generalization, and §7 outlines open challenges and future directions.

3 Training for Generalization

Training-based methods update model parameters to improve reasoning generalization under shifts in surface form, composition, length, difficulty, domain, or language. The central challenge is that training can reinforce any pattern that fits the training data, including shortcuts that break under shift (Dziri et al., 2023; Mirzadeh et al., 2025). We therefore organize these methods by training stage and, within each stage, by the component they modify. Appendix D provides the training-data inventory (Table 9), evaluation benchmarks (Table 13), and survey-wide evaluation metrics (Table 15); the training protocols and reported results are documented in Appendix D.3.

3.1 Pre-Training

At the earliest stage, pre-training provides the model with its initial knowledge and reasoning primitives (i.e., basic operations). By the component they modify, methods fall into data curation and optimizer design. Data curation controls what the corpus contains, covering the primitives that later compositions require (Zhang et al., 2025a) and raising the ratio of inferred to stated facts so that the model acquires rules rather than memorizing (Wang et al., 2024a; Abramov et al., 2025). Optimizer design instead changes how the model fits the same corpus, either by aligning gradients across data sources (Chen et al., 2026a) or by using small initialization and strong weight decay to favor primitives over memorization (Zhang et al., 2024d, 2025d), although generalizing solutions reached by training far beyond overfitting can transfer poorly to new knowledge (He et al., 2026).

3.2 Mid-Training

Subsequently, mid-training inserts a stage between pre-training and task-specific adaptation, which narrows the gap between the pre-training and post-training distributions and supplies skills required by the target tasks; without it, a model lacking such a skill tends to memorize the tasks during post-training rather than acquire it (Zhang et al., 2025a; Jin et al., 2025). By the objective of this stage, we distinguish two common approaches: (1) **Continued Pre-Training** methods, which apply next-token prediction to raw text from a domain under-represented in pre-training, directly narrowing this gap (Shao et al., 2024; Fesser et al., 2026).

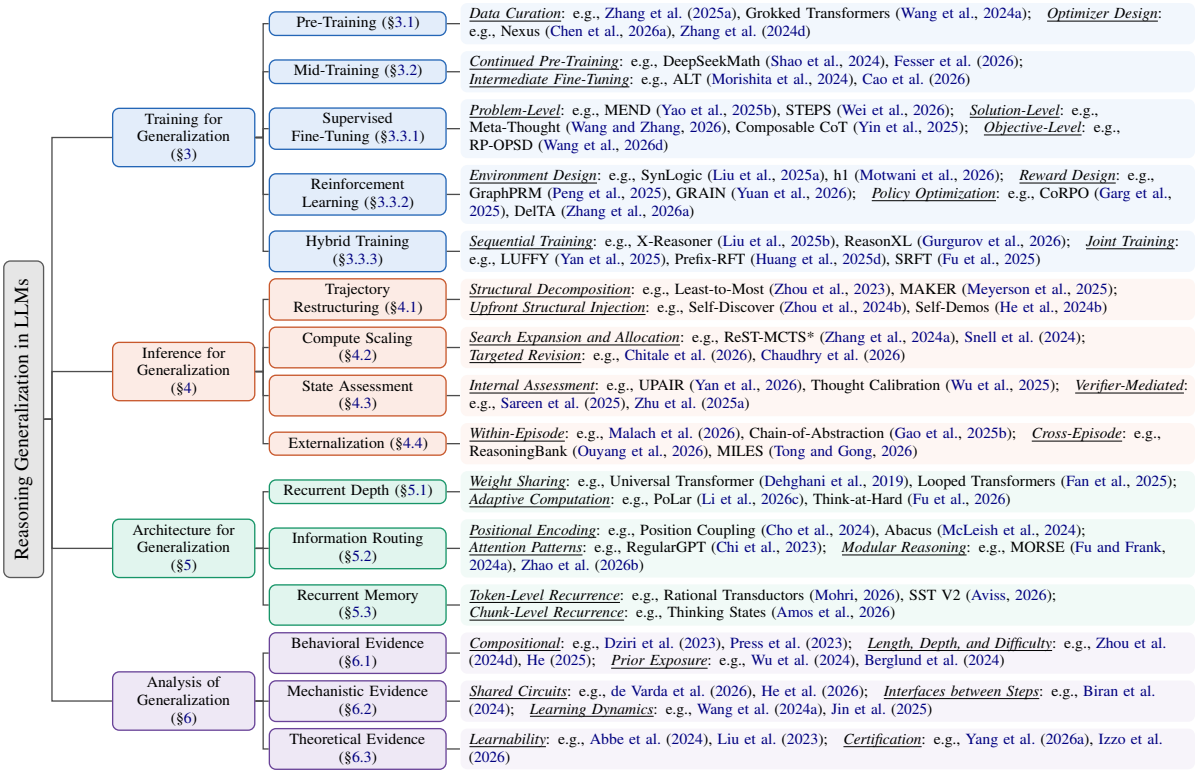


Figure 1: Taxonomy of reasoning generalization in LLMs.

(2) **Intermediate Fine-Tuning** methods, which train on solved problems that require no domain knowledge, such as program-generated deduction or symbolic problems, so that the acquired reasoning transfers to other domains and to realistic problems in natural language (Morishita et al., 2024; Cao et al., 2026). Beyond such direct transfer, this stage also prepares later training, after which the model generalizes better to OOD tasks and composes separately learned skills into novel combinations (Shrestha et al., 2025; RRV et al., 2026; Cheng et al., 2025b).

3.3 Post-Training

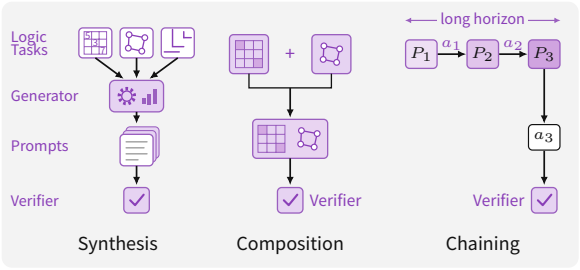
Post-training fine-tunes the model on reasoning tasks and is where the ID/OOD gap most easily opens: a model accurate on training tasks can still fail on rephrased, longer, or out-of-domain problems (Mirzadeh et al., 2025; Opedal et al., 2025; Gong et al., 2026). We cover Supervised Fine-Tuning (SFT), Reinforcement Learning (RL), and Hybrid Training.

3.3.1 Supervised Fine-Tuning

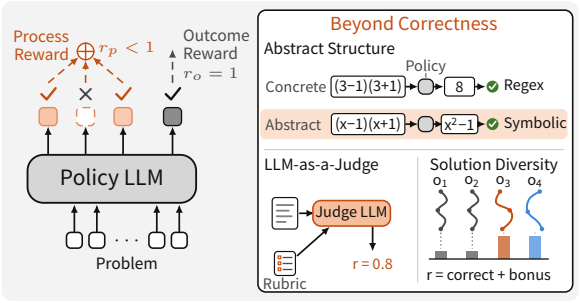
SFT trains the model to imitate demonstrations, i.e., step-by-step solutions to training problems, and thus tends to memorize their spurious surface correlations, becoming brittle under shift (Chu et al., 2025; Peng et al., 2026). Accordingly, SFT methods intervene at the level of the problems, the solutions, or the objective.

Problem-Level Methods in SFT. To begin with, this line of work modifies the training problems so that the shift anticipated at test time is already represented in training, leaving no single surface pattern for the model to memorize. (i) **Problem Rewriting** presents each problem in variants that differ only in solution-irrelevant details, teaching the model that the answer stays fixed under surface shift (Yao et al., 2025b; Gui and Ji, 2025). (ii) **Problem Generation** creates problems for regions the original data does not cover, such as rare skill combinations or slightly harder instances, which strengthens compositional and length generalization (Wei et al., 2026; Lee et al., 2025). (iii) **Difficulty Curriculum** orders or selects training problems by difficulty, generalizing to harder or OOD problems (Liu et al., 2026b,d); its benefit nevertheless depends on the shift, as difficulty ordering shows no robust advantage over random sampling for compositional generalization (Mordig et al., 2026).

Solution-Level Methods in SFT. In contrast, these methods rewrite the solutions that the model imitates. The central insight is that imitation gen-



(a) Environment Design



(b) Reward Design

Figure 2: Environment and reward design in RL: (a) building verifiable training problems, and (b) process, outcome, and beyond-correctness rewards.

eralizes only as far as its target does, so only a target exposing a reusable procedure transfers to new problems (Peng et al., 2026; Matsutani et al., 2026). (i) **Abstract Solutions** remove problem-specific details and teach a general strategy or rule before applying it to a specific problem, addressing surface and domain shift (Wang and Zhang, 2026; Peng et al., 2026). (ii) **Stepwise Solutions** write out every step of a rule allowing it to run for more steps or combine with others and thus to cope with length and compositional shift (Hu et al., 2024; Wang et al., 2026a; Hu et al., 2025b; Yin et al., 2025; Hua et al., 2025).

Objective-Level Methods in SFT. Inspired by knowledge distillation (Hinton et al., 2015) and invariance-based regularization from domain generalization (Arjovsky et al., 2019), these methods modify the loss so that what would not hold under shift contributes less to the update. (i) **Teacher-based Objectives** match the output distribution of a teacher on the student's own solutions instead of imitating a single written solution, which removes the mismatch with the expert solutions seen in training and enables transfer across domains, reasoning horizons, and languages (Li et al., 2026b; Wang et al., 2026d). More selective variants restrict this matching to tokens or spans the student can support, which yields further OOD gains (Jukić and Titov, 2026; Zhao et al., 2026c). (ii) **Regularization-based Objectives** constrain or reweight the imitation loss, keeping only gradient directions shared by logically equivalent problems or reweighting samples, including wrong-answer ones, and outperform standard SFT on OOD problems (Cheng et al., 2026; Tian et al., 2026). Despite these gains, a teacher given rich privileged information can suppress the student's uncertainty and degrade accuracy on unseen problems (Kim et al., 2026).

3.3.2 Reinforcement Learning

Unlike SFT, RL replaces reference solutions with a reward, usually whether a checker accepts the final answer, and updates the model toward higher-reward samples (Shao et al., 2024). Without imitation, RL reportedly generalizes to problem variants and domains where SFT degrades (Chu et al., 2025; Huan et al., 2025). Yet RL generalization is bounded: OOD gains seem to appear only when the target task aligns with pre-training (Ni et al., 2025), and accuracy collapses at a few times the training depth (Wang et al., 2026c). We further group RL methods by the component they modify: the environment, the reward, or the policy update (Figures 2 and 3).

Environment Design. First, these methods change the problems and checkers that the model practices on (Figure 2a) and hence control which tasks, difficulties, and reasoning horizons training covers. (i) **Synthesis** employs generators that turn logic tasks into checkable problems of controlled difficulty, with gains that extend to OOD puzzles and other domains (Liu et al., 2025a; Chen et al., 2025a); along this line, even environments built on fictional knowledge or games transfer to real-world problems (Kabra et al., 2026; Feng et al., 2026). (ii) **Composition** merges existing environments into tasks that require several skills at once, generalizing to unseen benchmarks (Xiang et al., 2026). (iii) **Chaining** feeds the answer of each short problem into the next, letting a single verifiable answer span a longer horizon and carry over to longer and harder problems (Motwani et al., 2026).

Reward Design. Second, instead of changing the problems, another family changes what is rewarded (Figure 2b). The central insight is that an outcome-only reward credits any path to a correct answer, including shortcuts that hold only in distribution; thus, this family also rewards properties of the reasoning itself. (i) **Process Rewards** score

each intermediate step (r_p), not merely the final answer judged by outcome rewards (r_o), favoring a correct procedure over such shortcuts (Salimi et al., 2026; Peng et al., 2025). (ii) **Rewards beyond Correctness** score properties of a solution other than its answer, in three forms. *Abstract structure* rewards favor solutions that capture the problem symbolically regardless of its names and narrative, narrowing the OOD gap of SFT and increasing robustness to surface perturbations (Yuan et al., 2026; Gao et al., 2025a). An *LLM-as-a-judge* replaces or complements rule-based checkers, extending rewards to answers that rules cannot verify (Ma et al., 2025b; Bhattarai et al., 2026) and raising accuracy outside the training domain (Zha et al., 2025; Ma et al., 2025b; Bhattarai et al., 2026). Lastly, *solution diversity* rewards grant a bonus to solutions that differ from the rest of the group, yielding higher out-of-domain accuracy (Hu et al., 2026b; Zhou et al., 2025; Maniparambil et al., 2026).

Policy Optimization. Finally, rather than changing the problems or the reward, these methods modify the policy update itself, so that it does not reinforce shortcuts that hold only on the training data. Figure 3 illustrates three such modifications to the standard GRPO update (Shao et al., 2024). (i) **Advantage Modification** reshapes the advantage, i.e., how much a rollout outscores its group, ensuring that incorrect rollouts with partial credit never receive a positive update, strengthening OOD transfer (Garg et al., 2025). (ii) **Token Weighting** assigns each token its own credit according to how strongly it signals a correct rather than an incorrect rollout (Zhang et al., 2026a; Xu et al., 2026b), avoiding the OOD degradation of standard updates (Xu et al., 2026b). (iii) **Gradient Regularization** penalizes the gradient norm to steer training toward flatter regions where the reward is more accurate, which curbs reward hacking, i.e., exploiting flaws of the reward (Ackermann et al., 2026).

3.3.3 Hybrid Training

Alternatively, hybrid methods combine demonstrations that introduce procedures the model cannot learn on its own with rewards that recover OOD accuracy reduced during SFT (Jin et al., 2025; Fesser et al., 2026). Depending on how the two signals are scheduled, we distinguish (i) **Sequential Training**, i.e., SFT followed by RL, which transfers reasoning to other domains, modalities, and languages (Liu et al., 2025b; Gurgurov et al., 2026); since

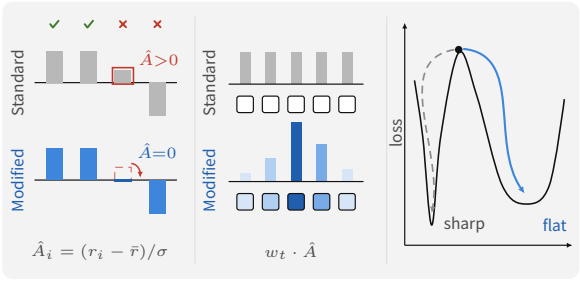


Figure 3: Policy optimization in RL: standard GRPO updates (gray) versus modified ones (blue).

excessive SFT leaves RL unable to improve, Rejuvenation (Liu et al., 2026c) partially resets the model toward its base weights before RL, restoring OOD gains. (ii) **Joint Training**, which combines the two signals within one stage, in the sampled solutions (Yan et al., 2025; Huang et al., 2025d), in the loss (Fu et al., 2025; Hu et al., 2026a), or by gating SFT and RL updates on rollout success (Lv et al., 2025; Ma et al., 2025a; Zhu et al., 2026), with several methods reporting higher OOD accuracy than RL alone (Yan et al., 2025; Fu et al., 2025; Lv et al., 2025). This advantage is challenged by Limozin et al. (2026): with corrected baselines, sequential training surpasses these joint methods, including OOD (Appendix D.3, Table 10).

4 Inference for Generalization

Autoregressive generation can compound errors on OOD tasks. Test-time methods target different parts of inference: structuring trajectories (Trajectory Restructuring), exploring alternatives (Compute Scaling), assessing reasoning reliability (State Assessment), and using external state or tools (Inference-Time Externalization). Their gains remain bounded by model capabilities: inference scaling cannot synthesize algorithmic reasoning entirely absent from training (Tong et al., 2026). Table 14 summarizes their evidence and caveats; Tables 13 and 15 list the datasets and metrics used throughout the survey.

4.1 Trajectory Restructuring

These methods organize free-form generation without updating model weights. **Structural decomposition** breaks complex reasoning into manageable steps (Zhou et al., 2023; Zhang et al., 2024c), using independently executed micro-agents (Meyerson et al., 2025) or predefined atomic skills (Chen

et al., 2024). **Upfront structural injection** constrains generation in advance by restricting valid answers (Ma et al., 2026), constructing task-specific procedures (Zhou et al., 2024b), or synthesizing query-specific demonstrations (He et al., 2024b). The latter addresses out-of-demonstration generalization, which should not be equated with an explicit shift from the pretraining distribution.

4.2 Compute Scaling

When one trajectory is insufficient, extra test-time compute can explore alternatives or revise errors. **Search expansion and allocation** uses process-level signals to guide exploration: ReST-MCTS* (Zhang et al., 2024a) combines a Process Reward Model (PRM) with Monte Carlo Tree Search (MCTS) to score partial solutions, while MCTS can also improve policy and reward models for finer inference guidance (Chan et al., 2025). Adaptive methods allocate compute according to estimated problem difficulty (Snell et al., 2024), although this alone does not establish OOD generalization. **Targeted revision** concentrates computation on faulty reasoning steps rather than regenerating an entire solution (Chitale et al., 2026).

4.3 State Assessment

State assessment estimates the reliability of intermediate reasoning to guide selection or further computation. **Internal assessment** draws on residual-stream trajectories (Damirchi et al., 2026), uncertainty-progress alignment (Yan et al., 2026), or thought calibration for risk-controlled stopping (Wu et al., 2025). Online calibration can further update reliability estimates under distribution shift (Zhou et al., 2026). **Verifier-mediated assessment** uses auxiliary models for candidate selection or adaptation: joint reasoner–verifier training supports inference-time selection in evaluated mathematical and physics tasks (Sareen et al., 2025), while verifier-guided test-time training adapts the generator using high-confidence outputs on unlabeled target data (Moradi et al., 2025). However, PRMs can generalize poorly to harder problems or solutions from different models (Zhu et al., 2025a).

4.4 Inference-Time Externalization

Externalization gives inference access to working state or computation beyond the model's internal state. **Within-episode externalization** uses interactive memory to support length extrapolation (Malach et al., 2026), separates abstract reasoning from tool-supplied values (Gao et al., 2025b), and supports tool selection or restraint (Alazraki and Rei, 2025; Qian et al., 2025). These gains do not by themselves establish that the model generalizes without external resources. **Cross-episode externalization** stores strategies distilled from prior successes and failures in persistent memory (Ouyang et al., 2026); modular sub-instructions and learned selection mechanisms support memory retrieval (Tong and Gong, 2026), although cross-domain transfer is not directly established by the main evaluations.

5 Architecture for Generalization

Architectural approaches change the computation available to a model: how it reuses transformations, connects representations, or carries state forward. Such changes provide inductive biases that can support reasoning generalization by allowing a model to reuse learned operations when problems grow longer or combine familiar steps in new ways.

5.1 Recurrent Depth

Recurrent depth repeatedly applies a shared transformation, increasing computation without a separate parameter set at each depth (Geiping et al., 2025; Zeng et al., 2026; Zhu et al., 2025c). This can support length generalization when longer problems require more iterations of the same operation, as shown on controlled algorithmic tasks (Fan et al., 2025). Designs differ in which computation is reused and how its iteration count is adjusted across inputs or tokens. **Weight Sharing.** Shared recurrent blocks support generalization to deeper compositions of facts (Kohli et al., 2026). On computation graphs, recurrence combined with latent supervision, discretization, and error correction enables generalization to larger instances (Altabaa et al., 2025). In pretrained language models, a shared middle block supports depth extrapolation on controlled rule-following tasks, although zero-shot transfer to verbal formulations remains limited (Shapiro, 2026). Looped language models also improve compositional tool calling involving dependent calls (Popescu et al., 2026). **Adaptive Computation.** PoLar (Li et al., 2026c) learns input-dependent programs that skip or repeat groups of layers in a frozen backbone. Think-at-Hard (Fu et al., 2026) selects tokens for extra latent iterations, and its math-only variant also improves on MMLU-STEM. Extra iterations can still degrade predic-

tions when recurrent states drift, limiting depth extrapolation (Viakhirev et al., 2026).

5.2 Information Routing

Information routing determines how tokens are addressed, which dependencies attention can access, and which components process them (Vaswani et al., 2017; Shazeer et al., 2017). For reasoning generalization, these designs aim to preserve the connections needed by a computation as input length or composition changes. **Positional Encoding.** Position Coupling (Cho et al., 2024) assigns shared position identifiers to digits of equal significance across operands, while Abacus embeddings (McLeish et al., 2024) encode digit positions within each number. Structural symmetry and alternative positional representations also improve arithmetic extrapolation (Sabbaghi et al., 2024; Shen et al., 2023). Multi-level position coupling (Cho et al., 2025) combines multiple position coordinates with task-specific scratchpads to generalize across operand lengths and counts. TAPE (Zhu et al., 2025b) instead updates positional representations across layers using sequence context, improving arithmetic length extrapolation. Much of this arithmetic evidence comes from specialized models. **Attention Patterns.** RegularGPT (Chi et al., 2023) combines sliding-dilated attention with shared layers and length-dependent depth, composing local results over larger spans to recognize regular languages beyond training lengths. Attention Bias Calibration (Duan et al., 2023) converts attention patterns learned on shorter inputs into biases that support arithmetic length extrapolation. **Modular Reasoning.** MORSE (Fu and Frank, 2024a) learns input-dependent masks that route tokens to specialized attention heads, improving generalization across entailment-tree lengths and shapes. In sparse expert language models, more complex skill compositions benefit from more active experts, while excessive activation can hurt simpler compositions (Zhao et al., 2026b).

5.3 Recurrent Memory

Recurrent memory carries an internal state across computation steps, allowing later steps to reuse earlier information (Bulatov et al., 2022). Updates occur across tokens or reasoning chunks, and generalization depends on retaining the needed information as sequences grow. **Token-Level Recurrence.** Rational Transductors (Mohri, 2026) combine weighted-automaton recurrence with transformer attention, providing theoretical and synthetic-task evidence for length generalization. SST V2 (Aviss, 2026) blends a carried state into pretrained feed-forward layers and reports gains beyond math fine-tuning, although its aggregate scores count a question as solved if any tested iteration depth succeeds. **Chunk-Level Recurrence.** Thinking States (Amos et al., 2026) compresses intermediate thoughts into a fixed-size state for later chunks, improving length extrapolation on parity and variable-tracking tasks, although the model still accesses preceding context alongside this state.

6 Analysis of Generalization

While the methods in §3–§5 aim to improve performance under distribution shift, analytical studies examine what transfers and why. We organize this work by three complementary types of evidence: behavioral evidence measures performance under controlled shifts; mechanistic evidence probes internal computation, mostly in small models; and theoretical evidence characterizes what can be represented, learned, and certified. Table 6 and Appendix C summarize each line of analysis.

6.1 Behavioral Evidence

Compositional generalization. Accuracy declines as computation graphs grow deeper and wider (Dziri et al., 2023), models can fail to chain two facts they both know (Zhang et al., 2026b), and this gap does not shrink with scale in the GPT-3 family (Press et al., 2023). **Length, depth, and difficulty.** Transformers fail on nesting deeper than seen in training (He, 2025), short-to-long transfer varies across random seeds (Zhou et al., 2024d), and training only on easy or only on hard examples yields inconsistent gains (Kordi et al., 2026). **Prior exposure.** Whether a test is "unseen" depends on the training stage it is compared against: accuracy drops on familiar tasks posed in an unfamiliar number base (Wu et al., 2024), facts learned as "A is B" are not retrieved as "B is A" (Berglund et al., 2024), and RL transfer requires sufficient base-model competence (Lu et al., 2025).

6.2 Mechanistic Evidence

Mechanistic studies reveal how internal computations shape generalization. **Shared circuits.** If two problem variants are solved by the same procedure, their internal computations should overlap. de Varda et al. (2026) show that, in LLM arithmetic, the overlap between circuits for numeric

and verbal formats predicts which formats are hardest, which models transfer best, and which items are solved. However, such a circuit can transfer poorly once it must use newly integrated knowledge (He et al., 2026). **Interfaces between steps.** Multi-step reasoning can also fail when an intermediate result does not reach the next step. Biran et al. (2024) find that LLMs resolve the bridge entity of a two-hop query in early layers but begin the second hop in later layers that may lack the required knowledge; patching a later hidden state into an earlier layer, chosen per query, recovers up to 66% of failed queries. **Learning dynamics.** When several solutions fit the training data, optimization determines which one the model reaches. Grokked transformers generalize OOD on comparison but not on composition (Wang et al., 2024a). In LLM post-training, OOD performance can peak early in SFT and then decline while ID performance still improves (Jin et al., 2025), making the evaluated checkpoint part of the generalization claim.

6.3 Theoretical Evidence

Theoretical work distinguishes three properties that empirical results often conflate: *expressivity*, whether a model class can represent a procedure that generalizes in length or composition; *learnability*, whether finite training selects this procedure over shortcuts that fit the same data; and *certification*, whether a finite test can establish that it extrapolates. Each result holds only under its stated encoding, precision, and computation budget. **Learnability.** Two conditions govern whether training selects the extrapolating procedure. The first is locality: Abbe et al. (2024) show, theoretically under additional assumptions and empirically, that targets with a high globality degree are not efficiently learnable by regular transformers, whereas inductive scratchpads, which condition each step only on the question and the previous state, break this barrier and improve length transfer. The second is inductive bias: Liu et al. (2023) prove that $O(\log T)$ layers suffice to simulate any finite automaton on inputs of length T, yet standard training in their experiments finds shortcuts that are brittle outside the training distribution. **Certification.** Yang et al. (2026a) prove that no computable length generalization bound exists for C-RASP, even with two layers, so no computable test length certifies the class. Positive results hold only for narrower classes: one- and two-layer transformers provably length-generalize when behavior on longer sequences can be simulated by behavior on shorter training sequences (Izzo et al., 2026).

7 Challenges and Future Directions

Challenges (1) **Limited transfer.** Models that perform well on familiar tasks can fail when known reasoning steps are combined in new ways or applied to different domains (§6.1). Learning procedures that remain effective under these changes is still a major challenge. (2) **Length and depth extrapolation.** Accuracy can decline when problems require more reasoning steps than those seen during training (Wang et al., 2026c). Additional computation at inference does not always resolve this difficulty, as recurrent states may drift (Viakhirev et al., 2026). (3) **Interpretability.** Behavioral scores alone do not show whether a model reuses the same reasoning procedure across tasks. Many mechanistic studies use small models, and their findings may not fully explain transfer in frontier LLMs (§6.2).

Future Directions (1) **Transferable training.** Developing training methods that teach reusable operations is central to improving generalization. Supervision should emphasize the reasoning steps that remain useful when familiar operations are combined in new ways (§3). (2) **Stable recurrent architectures.** Developing recurrent architectures with stable state updates may improve reasoning on longer and deeper problems. These architectures should maintain reliable computation as the number of reasoning steps increases (§5). (3) **Interpretability and verification.** Developing methods to identify and verify reusable computations is important for understanding reasoning generalization. Applying these methods to larger models would help connect mechanistic findings with the failures observed under distribution shift (§6.2).

8 Conclusion

This survey reviews reasoning generalization in LLMs from four perspectives: training, inference, architecture, and analysis. Existing studies show that generalization depends on what models learn, the computation available at inference, and the type of distribution shift. Although current methods achieve promising results in specific settings, none consistently generalizes across different shifts. Future research should develop clearer evaluation protocols and better explain when and why reasoning generalizes.

Limitations

While this survey offers a comprehensive review of existing methods and analyses of reasoning generalization in LLMs, a few limitations remain in literature coverage, comparison across studies, and applicability to larger models. Given the broad scope and rapid growth of this field, our review may have missed some relevant studies. Our focus on explicit reasoning-related distribution shifts limits its coverage of related work and leaves relatively few inference studies. Differences in models, tasks, metrics, and compute restrict the synthesis to qualitative comparisons. Many mechanistic and theoretical findings rely on small models or idealized assumptions, so their relevance to larger models remains uncertain. Finally, our taxonomy assigns each study a primary pillar, although some methods span several.

Ethical considerations

This work surveys publicly available research on reasoning generalization in LLMs and does not involve human participants, collect new personal data, or train or release new models. Reasoning methods may be used for both beneficial and harmful purposes, and improved transfer on selected benchmarks does not establish that a model is safe or reliable in deployment. We therefore encourage assessing generalization alongside bias, privacy, and misuse risks, especially in applications where reasoning errors can harm users.

References

Emmanuel Abbe, Samy Bengio, Aryo Lotfi, Colin Sandon, and Omid Saremi. 2024. How far can transformers reason? The globality barrier and inductive scratchpad. In *Advances in Neural Information Processing Systems*, volume 37.

Samira Abnar, Omid Saremi, Laurent Dinh, Shantel Wilson, Miguel Angel Bautista, Chen Huang, Vimal Thilak, Etai Littwin, Jiatao Gu, Josh Susskind, and Samy Bengio. 2023. Adaptivity and modularity for efficient generalization over task complexity. *Preprint*, arXiv:2310.08866.

Roman Abramov, Felix Steinbauer, and Gjergji Kasneci. 2025. Grokking in the wild: Data augmentation for real-world multi-hop reasoning with transformers. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*. PMLR.

Johannes Ackermann, Michael Noukhovitch, Takashi Ishida, and Masashi Sugiyama. 2026. Gradient regularization mitigates reward hacking in reinforcement learning from human feedback and verifiable rewards. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*. PMLR.

Mohamed Aghzal, Erion Plaku, Gregory J. Stein, and Ziyu Yao. 2025. A survey on large language models for automated planning. *Preprint*, arXiv:2502.12435.

Kartik Ahuja and Amin Mansouri. 2024. On provable length and compositional generalization. *Preprint*, arXiv:2402.04875.

Lisa Alazraki and Marek Rei. 2025. Meta-reasoning improves tool use in large language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 7900–7912.

Alon Albalak, Duy Phung, Nathan Lile, Rafael Rafailov, Kanishk Gandhi, Louis Castricato, Anikait Singh, Chase Blagden, Violet Xiang, Dakota Mahan, and Nick Haber. 2025. Big-Math: A large-scale, high-quality math dataset for reinforcement learning in language models. *Preprint*, arXiv:2502.17387.

Sarwan Ali. 2026. Critical windows of complexity control: When transformers decide to reason or memorize. *Preprint*, arXiv:2605.04396.

Awni Altabaa, Siyu Chen, John Lafferty, and Zhuoran Yang. 2025. Unlocking out-of-distribution generalization in transformers via recursive latent space reasoning. *Preprint*, arXiv:2510.14095.

Ido Amos, Avi Caciularu, Mor Geva, Amir Globerson, Jonathan Herzig, Lior Shani, and Idan Szpektor. 2026. Latent reasoning with supervised thinking states. *Preprint*, arXiv:2602.08332.

Zhiyu An and Wan Du. 2026. Representational homomorphism predicts and improves compositional generalization in transformer language model. *Preprint*, arXiv:2601.18858.

Cem Anil, Yuhuai Wu, Anders Andreassen, Aitor Lewkowycz, Vedant Misra, Vinay Ramasesh, Ambrose Slone, Guy Gur-Ari, Ethan Dyer, and Behnam Neyshabur. 2022. Exploring length generalization in large language models. In *Advances in Neural Information Processing Systems*, volume 35.

Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. 2019. Invariant risk minimization. *Preprint*, arXiv:1907.02893.

Thea Aviss. 2026. State Stream Transformer (SST) V2: Parallel training of nonlinear recurrence for latent space reasoning. *Preprint*, arXiv:2605.00206.

Malia Barker, Bishal Lakha, Edoardo Serra, and Francesco Gullo. 2026. Testing LLM arithmetic reasoning generalization with automatic numeric-remapping attacks. *Preprint*, arXiv:2606.03606.

Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. 2024. The reversal curse: LLMs trained on "A is B" fail to learn "B is A". In *International Conference on Learning Representations*.

Manish Bhattarai, Ismael Boureima, Nishath Rajiv Ranasinghe, Scott Pakin, and Dan O'Malley. 2026. Rubric-grounded RL: Structured judge rewards for generalizable reasoning. *Preprint*, arXiv:2605.08061.

Eden Biran, Daniela Gottesman, Sohee Yang, Mor Geva, and Amir Globerson. 2024. Hopping too late: Exploring the limitations of large language models on multi-hop queries. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14113–14130.

Aydar Bulatov, Yury Kuratov, and Mikhail Burtsev. 2022. Recurrent memory transformer. In *Advances in Neural Information Processing Systems*, volume 35.

Ziyang Cai, Nayoung Lee, Avi Schwarzschild, Samet Oymak, and Dimitris Papailiopoulos. 2025. Extrapolation by association: Length generalization transfer in transformers. *Preprint*, arXiv:2506.09251.

Mingzi Cao, Xingwei Tan, Mahmud Elahi Akhter, Marco Valentino, Maria Liakata, Xi Wang, and Nikolaos Aletras. 2026. Fundamental reasoning paradigms induce out-of-domain generalization in language models. *Preprint*, arXiv:2602.08658.

Chi-Min Chan, Chunpu Xu, Junqi Zhu, Jiaming Ji, Donghai Hong, Pengcheng Wen, Chunyang Jiang, Zhen Ye, Yaodong Yang, Wei Xue, Sirui Han, and Yike Guo. 2025. Boosting policy and process reward models with Monte Carlo tree search in open-domain QA. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 7433–7451.

Hoyeon Chang, Jinho Park, Hanseul Cho, Sohee Yang, Miyoung Ko, Hyeonbin Hwang, Seungpil Won, Dohaeng Lee, Youbin Ahn, and Minjoon Seo. 2025. Characterizing pattern matching and its limits on compositional task structures. *Preprint*, arXiv:2505.20278.

Arslan Chaudhry, Sridhar Thiagarajan, and Andrew Lampinen. 2026. Improving latent generalization using test-time compute. *Preprint*, arXiv:2604.01430.

Huanran Chen, Huaqing Zhang, Xiao Li, Yinpeng Dong, Ke Shen, and Jun Zhu. 2026a. Nexus: Same pretraining loss, better downstream generalization via common minima. *Preprint*, arXiv:2604.09258.

Hung-Hsuan Chen. 2026. Thinking deeper, not longer: Memory-efficient test-time reasoning with depth-recurrent transformers for compositional generalization. *Preprint*, arXiv:2603.21676.

Jiaao Chen, Xiaoman Pan, Dian Yu, Kaiqiang Song, Xiaoyang Wang, Dong Yu, and Jianshu Chen. 2024. Skills-in-context: Unlocking compositionality in large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 13838–13890.

Jiangjie Chen, Qianyu He, Siyu Yuan, Aili Chen, Zhicheng Cai, Weinan Dai, Hongli Yu, Qiying Yu, Xuefeng Li, Jiaze Chen, Hao Zhou, and Mingxuan Wang. 2025a. Enigmata: Scaling logical reasoning in large language models with synthetic verifiable puzzles. *Preprint*, arXiv:2505.19914.

Kedi Chen, Dezhao Ruan, Yuhao Dan, Yaoting Wang, Siyu Yan, Xuecheng Wu, Yinqi Zhang, Qin Chen, Jie Zhou, Liang He, Biqing Qi, Linyang Li, Qipeng Guo, Xiaoming Shi, and Wei Zhang. 2025b. A survey of inductive reasoning for large language models. *Preprint*, arXiv:2510.10182.

Mark Chen et al. 2021. Evaluating large language models trained on code. *Preprint*, arXiv:2107.03374.

Michelle Chao Chen, Moritz Miller, Bernhard Schölkopf, and Siyuan Guo. 2026b. On the emergence and test-time use of structural information in large language models. *Preprint*, arXiv:2601.17869.

Xinghao Chen, Anhao Zhao, Heming Xia, Xuan Lu, Hanlin Wang, Yanjun Chen, Wei Zhang, Jian Wang, Wenjie Li, and Xiaoyu Shen. 2026c. Reasoning beyond language: A comprehensive survey on latent chain-of-thought reasoning. *Preprint*, arXiv:2505.16782.

Fengxiang Cheng, Haoxuan Li, Fenrong Liu, Robert van Rooij, Kun Zhang, and Zhouchen Lin. 2025a. Empowering LLMs with logical reasoning: A comprehensive survey. *Preprint*, arXiv:2502.15652.

Sitao Cheng, Xunjian Yin, Ruiwen Zhou, Yuxuan Li, Xinyi Wang, Liangming Pan, William Yang Wang, and Victor Zhong. 2025b. Atomic skills are the prerequisite: When reinforcement learning synthesizes compositional reasoning, and when it only amplifies. *Preprint*, arXiv:2512.01970.

Zehua Cheng, Wei Dai, and Jiahao Sun. 2026. Invariant gradient alignment for robust reasoning distillation. *Preprint*, arXiv:2606.05025.

Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, Yutao Xie, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jianshu She, Chengqian Gao, Abulhair Saparov, Haonan Li, and 5 others. 2025c. Revisiting reinforcement learning for LLM reasoning from a cross-domain perspective. *Preprint*, arXiv:2506.14965.

Nathan A. Chi, Teodor Malchev, Riley Kong, Ryan A. Chi, Lucas Huang, Ethan A. Chi, R. Thomas McCoy, and Dragomir Radev. 2024. modeLing: A novel dataset for testing linguistic reasoning in language models. *Preprint*, arXiv:2406.17038.

Ta-Chung Chi, Ting-Han Fan, Alexander Rudnicky, and Peter Ramadge. 2023. Transformer working memory enables regular language reasoning and natural language length extrapolation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5972–5984, Singapore. Association for Computational Linguistics.

Rajiv Shailesh Chitale, Rahul Madhavan, Taneesh Gupta, Deepanway Ghosal, and Aravindan Raghuveer. 2026. Test-time scaling via error localization. *Preprint*, arXiv:2607.21453.

Hanseul Cho, Jaeyoung Cha, Pranjal Awasthi, Srinadh Bhojanapalli, Anupam Gupta, and Chulhee Yun. 2024. Position coupling: Improving length generalization of arithmetic transformers using task structure. In *Advances in Neural Information Processing Systems*, volume 37, pages 22233–22315.

Hanseul Cho, Jaeyoung Cha, Srinadh Bhojanapalli, and Chulhee Yun. 2025. Arithmetic transformers can length-generalize in both operand length and count. In *International Conference on Learning Representations*.

François Chollet. 2019. On the measure of intelligence. *Preprint*, arXiv:1911.01547.

Moumita Choudhury, Vanshaj Khattar, Jing Liu, Toshiaki Koike-Akino, Ankush Chakrabarty, Shlomo Zilberstein, and Ye Wang. 2026. Training language models to cooperate with inference-time controllers. *Preprint*, arXiv:2607.23771.

Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, and Yi Ma. 2025. SFT memorizes, RL generalizes: A comparative study of foundation model post-training. *Preprint*, arXiv:2501.17161.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. *Preprint*, arXiv:1803.05457.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *Preprint*, arXiv:2110.14168.

Lu Dai, Ziyang Rao, Yili Wang, Hanqing Wang, Hao Liu, and Hui Xiong. 2026. Towards mechanistically understanding why memorized knowledge fails to generalize in large language model finetuning. *Preprint*, arXiv:2607.08393.

Hamed Damirchi, Ignacio Meza De la Jara, Damith Ranasinghe, Yuhang Liu, and Javen Shi. 2026. Reasoning errors have a region and a direction in the residual-stream trajectory of LLMs. *Preprint*, arXiv:2608.05660.

Andrea Gregor de Varda, Sana Pandey, Pengrui Han, Jacob Andreas, and Evelina Fedorenko. 2026. Shared circuits predict whether LLMs generalize across formats in arithmetic reasoning. *Preprint*, arXiv:2609.04463.

Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. 2019. Universal transformers. In *International Conference on Learning Representations*.

Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. 2023. Mind2Web: Towards a generalist agent for the web. In *Advances in Neural Information Processing Systems*, volume 36, pages 28091–28114.

Mucong Ding, Chenghao Deng, Jocelyn Choo, Zichu Wu, Aakriti Agrawal, Avi Schwarzschild, Tianyi Zhou, Tom Goldstein, John Langford, Anima Anandkumar, and Furong Huang. 2024. Easy2Hard-Bench: Standardized difficulty labels for profiling LLM performance and generalization. In *Advances in Neural Information Processing Systems*, volume 37.

Xinrun Du et al. 2025. SuperGPQA: Scaling LLM evaluation across 285 graduate disciplines. *Preprint*, arXiv:2502.14739.

Shaoxiong Duan, Yining Shi, and Wei Xu. 2023. From interpolation to extrapolation: Complete length generalization for arithmetic transformers. *Preprint*, arXiv:2310.11984.

Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D. Hwang, Soumya Sanyal, Sean Welleck, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. 2023. Faith and fate: Limits of transformers on compositionality. In *Advances in Neural Information Processing Systems*, volume 36.

Ying Fan, Yilun Du, Kannan Ramchandran, and Kangwook Lee. 2025. Looped transformers for length generalization. In *International Conference on Learning Representations*.

Xiachong Feng, Deyi Yin, Xiaocheng Feng, Yi Jiang, Libo Qin, Yangfan Ye, Lei Huang, Weitao Ma, Qiming Li, Yuxuan Gu, Bing Qin, and Lingpeng Kong. 2026. Stratagem: Learning transferable reasoning via trajectory-modulated game self-play. *Preprint*, arXiv:2604.17696.

Lukas Fesser, Hanlin Zhang, Michelle M. Li, Eric Wang, Bryan Perozzi, Shekoofeh Azizi, Sham M. Kakade, and Marinka Zitnik. 2026. How post-training shapes biological reasoning models. *Preprint*, arXiv:2606.16517.

Tianyu Fu, Yichen You, Zekai Chen, Guohao Dai, Huazhong Yang, and Yu Wang. 2026. Think-at-hard: Dynamic looped transformers for improved reasoning. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of

Proceedings of Machine Learning Research, pages 32129–32154. PMLR.

Xiyan Fu and Anette Frank. 2024a. Compositional structured explanation generation with dynamic modularized reasoning. In *Proceedings of the 13th Joint Conference on Lexical and Computational Semantics (*SEM 2024)*, pages 385–401, Mexico City, Mexico. Association for Computational Linguistics.

Xiyan Fu and Anette Frank. 2024b. The mystery of compositional generalization in graph-based generative commonsense reasoning. *Preprint*, arXiv:2410.06272.

Yuqian Fu, Tinghong Chen, Jiajun Chai, Xihuai Wang, Songjun Tu, Guojun Yin, Wei Lin, Qichao Zhang, Yuanheng Zhu, and Dongbin Zhao. 2025. SRFT: A single-stage method with supervised and reinforcement fine-tuning for reasoning. *Preprint*, arXiv:2506.19767.

GAIR. 2025. LIMO released training dataset. Hugging Face dataset.

Silin Gao, Antoine Bosselut, Samy Bengio, and Emmanuel Abbe. 2025a. AbstRaL: Augmenting LLMs' reasoning by reinforcing abstract thinking. *Preprint*, arXiv:2506.07751.

Silin Gao, Jane Dwivedi-Yu, Ping Yu, Xiaoqing Ellen Tan, Ramakanth Pasunuru, Olga Golovneva, Koustuv Sinha, Asli Celikyilmaz, Antoine Bosselut, and Tianlu Wang. 2025b. Efficient tool use with chain-of-abstraction reasoning. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 2727–2743.

Anisha Garg, Claire Zhang, Nishit Neema, David Bick, Ganesh Venkatesh, and Joel Hestness. 2025. CoRPO: Adding a correctness bias to GRPO improves generalization. *Preprint*, arXiv:2511.04439.

Jonas Geiping, Sean McLeish, Neel Jain, John Kirchenbauer, Siddharth Singh, Brian Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, and Tom Goldstein. 2025. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In *Advances in Neural Information Processing Systems*, volume 38, pages 41340–41391.

Zhiren Gong, Tiantong Wu, Jiaming Zhang, Fuyao Zhang, Che Wang, Yurong Hao, Yikun Hou, Foo Ping, Yilei Zhao, Fei Huang, Chau Yuen, and Wei Yang Bryan Lim. 2026. XDomainBench: Diagnosing reasoning collapse in high-dimensional scientific knowledge composition. *Preprint*, arXiv:2605.14754.

Wenjie Guan and Jelena Bradic. 2026. When symbol names should not matter: A logistic theory of fresh-symbol classification. *Preprint*, arXiv:2605.07120.

Etash Guha et al. 2025. OpenThoughts: Data recipes for reasoning models. *Preprint*, arXiv:2506.04178.

Shurui Gui and Shuiwang Ji. 2025. Mitigating spurious correlations in LLMs via causality-aware post-training. *Preprint*, arXiv:2506.09433.

Daniil Gurgurov, Tom Röhr, Sebastian von Rohrscheidt, Josef van Genabith, Alexander Löser, and Simon Ostermann. 2026. ReasonXL: Shifting LLM reasoning language without sacrificing performance. *Preprint*, arXiv:2604.12378.

Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. 2024a. OlympiadBench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal scientific problems. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3828–3850.

Jujie He, Jiacai Liu, Chris Yuhao Liu, Rui Yan, Chaojie Wang, Peng Cheng, Xiaoyu Zhang, Fuxiang Zhang, Jiacheng Xu, Wei Shen, Siyuan Li, Liang Zeng, Tianwen Wei, Cheng Cheng, Bo An, Yang Liu, and Yahui Zhou. 2025a. Skywork Open Reasoner 1 technical report. *Preprint*, arXiv:2505.22312.

Kaiyu He, Zhang Mian, Peilin Wu, Xinya Du, and Zhiyu Chen. 2026. Is grokking worthwhile? Functional analysis and transferability of generalization circuits in transformers. *Preprint*, arXiv:2601.09049.

Wei He, Shichun Liu, Jun Zhao, Yiwen Ding, Yi Lu, Zhiheng Xi, Tao Gui, Qi Zhang, and Xuanjing Huang. 2024b. Self-Demos: Eliciting out-of-demonstration generalizability in large language models. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 3829–3845.

Zhiwei He, Tian Liang, Jiahao Xu, Qiuzhi Liu, Xingyu Chen, Yue Wang, Linfeng Song, Dian Yu, Zhenwen Liang, Wenxuan Wang, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, Haitao Mi, and Dong Yu. 2025b. DeepMath-103K: A large-scale, challenging, decontaminated, and verifiable mathematical dataset for advancing reasoning. *Preprint*, arXiv:2504.11456.

Zhiyuan He. 2025. Exploring depth generalization in large language models for solving recursive logic tasks. *Preprint*, arXiv:2512.02677.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. *Preprint*, arXiv:1503.02531.

Kaihua Hou, Abhijith Varma Mudunuri, Jiaxing Qiu, Roxana Daneshjou, Thomas Hartvigsen, and Ahmed Alaa. 2026. Test-time hinting for black-box vision-language models. *Preprint*, arXiv:2605.16410.

Jingcheng Hu, Yinmin Zhang, Qi Han, Daxin Jiang, Xiangyu Zhang, and Heung-Yeung Shum. 2025a. Open-Reasoner-Zero: An open source approach to scaling up reinforcement learning on the base model. *Preprint*, arXiv:2503.24290.

Yi Hu, Shijia Kang, Haotong Yang, Haotian Xu, and Muhan Zhang. 2025b. Beyond single-task: Robust multi-task length generalization for LLMs. *Preprint*, arXiv:2502.11525.

Yi Hu, Xiaojuan Tang, Haotong Yang, and Muhan Zhang. 2024. Case-based or rule-based: How do transformers do the math? In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR.

Yuelin Hu, Zhenbo Yu, Zhengxue Cheng, Wei Liu, and Li Song. 2026a. GAC: Noise-aware adaptive mixing for hybrid SFT-RL post-training. *Preprint*, arXiv:2605.26184.

Zican Hu, Shilin Zhang, Yafu Li, Jianhao Yan, Xuyang Hu, Leyang Cui, Xiaoye Qu, Chunlin Chen, Yu Cheng, and Zhi Wang. 2026b. Diversity-incentivized exploration for versatile reasoning. In *International Conference on Learning Representations*.

Zhouqi Hua, Wenwei Zhang, Chengqi Lyu, Yuzhe Gu, Songyang Gao, Kuikun Liu, Dahua Lin, and Kai Chen. 2025. The imitation game: Turing machine imitator is length generalizable reasoner. *Preprint*, arXiv:2507.13332.

Maggie Huan, Yuetai Li, Tuney Zheng, Xiaoyu Xu, Seungone Kim, Minxin Du, Radha Poovendran, Graham Neubig, and Xiang Yue. 2025. Does math reasoning improve general LLM capabilities? Understanding transferability of LLM reasoning. *Preprint*, arXiv:2507.00432.

Benhao Huang, Zhengyang Geng, and J. Zico Kolter. 2026a. Equilibrium reasoners: Learning attractors enables scalable reasoning. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 46586–46616. PMLR.

Kaixuan Huang, Jiacheng Guo, Zihao Li, Xiang Ji, Jiawei Ge, Wenzhe Li, Yingqing Guo, Tianle Cai, Hui Yuan, Runzhe Wang, Yue Wu, Ming Yin, Shange Tang, Yangsibo Huang, Chi Jin, Xinyun Chen, Chiyuan Zhang, and Mengdi Wang. 2025a. MATH-Perturb: Benchmarking LLMs' math reasoning abilities against hard perturbations. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 25311–25328. PMLR.

Xinting Huang, Aleksandra Bakalova, Satwik Bhattamishra, William Merrill, and Michael Hahn. 2026b. Discovering interpretable algorithms by decompiling transformers to RASP. *Preprint*, arXiv:2602.08857.

Xinting Huang, Andy Yang, Satwik Bhattamishra, Yash Sarrof, Andreas Krebs, Hattie Zhou, Preetum Nakkiran, and Michael Hahn. 2025b. A formal framework for understanding length generalization in transformers. In *International Conference on Learning Representations*.

Yu Huang, Zixin Wen, Aarti Singh, Yuejie Chi, and Yuxin Chen. 2025c. Transformers provably learn chain-of-thought reasoning with length generalization. *Preprint*, arXiv:2511.07378.

Zeyu Huang, Tianhao Cheng, Zihan Qiu, Zili Wang, Yinghui Xu, Edoardo M. Ponti, and Ivan Titov. 2025d. Blending supervised and reinforcement fine-tuning with prefix sampling. *Preprint*, arXiv:2507.01679.

Hugging Face. 2025. Open R1: A fully open reproduction of DeepSeek-R1. `https://github.com/huggingface/open-r1`.

Dieuwke Hupkes, Mario Giulianelli, Verna Dankers, Mikel Artetxe, Yanai Elazar, Tiago Pimentel, Christos Christodoulopoulos, Karim Lasri, Naomi Saphra, Arabella Sinclair, Dennis Ulmer, Florian Schottmann, Khuyagbaatar Batsuren, Kaiser Sun, Koustuv Sinha, Leila Khalatbari, Maria Ryskina, Rita Frieske, Ryan Cotterell, and Zhijing Jin. 2023. A taxonomy and review of generalization research in NLP. *Nature Machine Intelligence*, 5(10):1161–1174.

Imran Ibrahimli, Stefan Wermter, and Jae Hee Lee. 2026. Adaptive recurrence as algorithmic time for length generalization in addition. In *2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents at ICML*.

Takuya Ito, Ruchir Puri, Murray Campbell, and Parikshit Ram. 2026. Universal transformers for circuit computations: Perfect length generalization in tiny transformers. *Preprint*, arXiv:2608.31067.

Zachary Izzo, Eshaan Nichani, and Jason D. Lee. 2026. Quantitative bounds for length generalization in transformers. In *International Conference on Learning Representations*.

Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. 2024. LiveCodeBench: Holistic and contamination free evaluation of large language models for code. *Preprint*, arXiv:2403.07974.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R. Narasimhan. 2024. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations*.

Hangzhan Jin, Sitao Luan, Tianwei Ni, Sicheng Lyu, Guillaume Rabusseau, Reihaneh Rabbany, Doina Precup, and Mohammad Hamdaqa. 2025. RL fine-tuning heals OOD forgetting in SFT. *Preprint*, arXiv:2509.12235.

Josip Jukić and Ivan Titov. 2026. Geometric self-distillation for reasoning generalization. *Preprint*, arXiv:2607.06855.

Anmol Kabra, Yilun Yin, Albert Gong, Kamilė Stankevičiūtė, Dongyoung Go, Johann Lee, Katie Z. Luo, Carla P. Gomes, and Kilian Q. Weinberger. 2026. Learning from synthetic data improves multi-hop reasoning. *Preprint*, arXiv:2603.02091.

Danial Kamali, Elham J. Barezi, and Parisa Kordjamshidi. 2025. NeSyCoCo: A neuro-symbolic concept composer for compositional generalization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 4184–4193.

Amirhossein Kazemnejad, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Payel Das, and Siva Reddy. 2023. The impact of positional encoding on length generalization in transformers. In *Advances in Neural Information Processing Systems*, volume 36.

Daniel Keysers, Nathanael Schärli, Nathan Scales, Hylke Buisman, Daniel Furrer, Sergii Kashubin, Nikola Momchev, Danila Sinopalnikov, Lukasz Stafiniak, Tibor Tihon, Dmitry Tsarkov, Xiao Wang, Marc van Zee, and Olivier Bousquet. 2020. Measuring compositional generalization: A comprehensive method on realistic data. In *International Conference on Learning Representations*.

Jeonghye Kim, Xufang Luo, Minbeom Kim, Sangmook Lee, Dohyung Kim, Jiwon Jeon, Dongsheng Li, and Yuqing Yang. 2026. Why does self-distillation (sometimes) degrade the reasoning capability of LLMs? *Preprint*, arXiv:2603.24472.

Najoung Kim and Tal Linzen. 2020. COGS: A compositional generalization challenge based on semantic interpretation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9087–9105.

Harsh Kohli, Srinivasan Parthasarathy, Huan Sun, and Yuekun Yao. 2026. Loop, think, & generalize: Implicit reasoning in recurrent-depth transformers. *Preprint*, arXiv:2604.07822.

Yeganeh Kordi, Nihal V. Nayak, Max Zuo, Ilana Nguyen, and Stephen H. Bach. 2026. Revisiting generalization across difficulty levels: It's not so easy. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7014–7042.

Oliver Kraus, Yash Sarrof, Yuekun Yao, Alexander Koller, and Michael Hahn. 2026. Barriers to universal reasoning with transformers (and how to overcome them). *Preprint*, arXiv:2604.25800.

Hsun-Yu Kuo, El Mahdi Chayti, Patrik Reizinger, Wieland Brendel, and Martin Jaggi. 2026. Stabilizing extrapolation in looped transformers via learned stochastic stopping. *Preprint*, arXiv:2606.29983.

Asher Labovich. 2026. Stability and generalization in looped transformers. *Preprint*, arXiv:2604.15259.

Brenden M. Lake and Marco Baroni. 2018. Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*. PMLR.

Nayoung Lee, Ziyang Cai, Avi Schwarzschild, Kangwook Lee, and Dimitris Papailiopoulos. 2025. Self-improving transformers overcome easy-to-hard and length generalization challenges. *Preprint*, arXiv:2502.01612.

Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam Neyshabur, Guy Gur-Ari, and Vedant Misra. 2022. Solving quantitative reasoning problems with language models. In *Advances in Neural Information Processing Systems*, volume 35.

Changhao Li, Yuchen Zhuang, Chenxiao Gao, Haotian Sun, Rushi Qiang, Chao Zhang, and Bo Dai. 2026a. Exploration-driven optimization for test-time large language model reasoning. *Preprint*, arXiv:2605.09853.

Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Costa Huang, Kashif Rasul, Longhui Yu, Albert Jiang, Ziju Shen, Zihan Qin, Bin Dong, Li Zhou, Yann Fleureau, Guillaume Lample, and Stanislas Polu. 2024a. NuminaMath: The largest public dataset in AI4Maths with 860k pairs of competition math problems and solutions. Hugging Face repository.

Qintong Li, Leyang Cui, Xueliang Zhao, Lingpeng Kong, and Wei Bi. 2024b. GSM-Plus: A comprehensive benchmark for evaluating the robustness of LLMs as mathematical problem solvers. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2961–2984.

Yunxin Li, Zhenyu Liu, Zitao Li, Xuanyu Zhang, Zhenran Xu, Xinyu Chen, Haoyuan Shi, Shenyuan Jiang, Xintong Wang, Jifang Wang, Shouzheng Huang, Xinping Zhao, Borui Jiang, Lanqing Hong, Longyue Wang, Zhuotao Tian, Baoxing Huai, Wenhan Luo, Weihua Luo, and 3 others. 2025. Perception, reason, think, and plan: A survey on large multimodal reasoning models. *Preprint*, arXiv:2505.04921.

Zhaoyi Li, Deyang Kong, Yuan Wei, Evan Yang, Ranran Shen, Mahardika Krisna Ihsani, Ming Yang, Wei Zhang, Chuan Hao, Jian Yang, Ran Tao, Bryan Dai, Shikun Zhang, Wei Ye, Ying Wei, and Defu Lian. 2026b. Every coin has two sides: On the dual nature of generalization in on-policy distillation of large language models. *Preprint*, arXiv:2608.16647.

Zhiyuan Li, Hong Liu, Denny Zhou, and Tengyu Ma. 2024c. Chain of thought empowers transformers to solve inherently serial problems. In *International Conference on Learning Representations*.

Ziyue Li, Yang Li, and Tianyi Zhou. 2026c. Skip a layer or loop it? Learning program-of-layers in LLMs. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 68114–68135. PMLR.

Hao Liang, Mingrui Chen, Hengyi Feng, Meiyi Qiang, and Wentao Zhang. 2026. DataFlex-RL: An evaluation platform for RLVR data policies. *Preprint*, arXiv:2609.06107.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. Let's verify step by step. In *International Conference on Learning Representations*.

Alexis Limozin, Eduard Durech, Torsten Hoefler, Imanol Schlag, and Valentina Pyatkin. 2026. SFT-then-RL outperforms mixed-policy methods for LLM reasoning. *Preprint*, arXiv:2604.23747.

Bill Yuchen Lin, Ronan Le Bras, Kyle Richardson, Ashish Sabharwal, Radha Poovendran, Peter Clark, and Yejin Choi. 2025. ZebraLogic: On the scaling limits of LLMs for logical reasoning. *Preprint*, arXiv:2502.01100.

Bingbin Liu, Jordan T. Ash, Surbhi Goel, Akshay Krishnamurthy, and Cyril Zhang. 2023. Transformers learn shortcuts to automata. In *International Conference on Learning Representations*.

Jiashuo Liu, Zheyan Shen, Yue He, Xingxuan Zhang, Renzhe Xu, Han Yu, and Peng Cui. 2021. Towards out-of-distribution generalization: A survey. *Preprint*, arXiv:2108.13624.

Junlin Liu, Shengnan An, Shuang Zhou, Dan Ma, Shixiong Luo, Ying Xie, Yuan Zhang, Wenling Yuan, Yifan Zhou, Xiaoyu Li, Ziwen Wang, Xuezhi Cao, and Xunliang Cai. 2026a. General365: Benchmarking general reasoning in large language models across diverse and challenging tasks. *Preprint*, arXiv:2604.11778.

Junteng Liu, Yuanxiang Fan, Zhuo Jiang, Han Ding, Yongyi Hu, Chi Zhang, Yiqi Shi, Shitong Weng, Aili Chen, Shiqi Chen, Yunan Huang, Mozhi Zhang, Pengyu Zhao, Junjie Yan, and Junxian He. 2025a. SynLogic: Synthesizing verifiable reasoning data at scale for learning logical reasoning and beyond. *Preprint*, arXiv:2505.19641.

Minghui Liu, Thomas Magelinski, Dehao Yuan, Qi Yu, and Furong Huang. 2026b. Ladders of thought: A self-evolving curriculum of progressively simplified reasoning traces. *Preprint*, arXiv:2609.25643.

Qianchu Liu, Sheng Zhang, Guanghui Qin, Timothy Ossowski, Yu Gu, Ying Jin, Sid Kiblawi, Sam Preston, Mu Wei, Paul Vozila, Tristan Naumann, and Hoifung Poon. 2025b. X-Reasoner: Towards generalizable reasoning across modalities and domains. *Preprint*, arXiv:2505.03981.

Runze Liu, Jiashun Liu, Xu Wan, Yuqian Fu, and Ling Pan. 2026c. When RL fails after SFT: Rejuvenating model plasticity for robust SFT-to-RL handoff. *Preprint*, arXiv:2606.09932.

Siyuan Liu, Tinghong Chen, Xinghan Li, Yifei Wang, and Jingzhao Zhang. 2026d. Data difficulty and the generalization–extrapolation tradeoff in LLM fine-tuning. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*. PMLR.

Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. 2025c. Understanding R1-Zero-like training: A critical perspective. *Preprint*, arXiv:2503.20783.

Brian Lu, Hongyu Zhao, Shuo Sun, Hao Peng, Rui Ding, and Hongyuan Mei. 2025. Generalization of RLVR using causal reasoning as a testbed. *Preprint*, arXiv:2512.20760.

Michael Luo, Sijun Tan, Justin Wong, Xiaoxiang Shi, William Y. Tang, Manan Roongta, Colin Cai, Jeffrey Luo, Li Erran Li, Raluca Ada Popa, and Ion Stoica. 2025. DeepScaleR: Surpassing O1-Preview with a 1.5B model by scaling RL. Notion blog.

Xingtai Lv, Yuxin Zuo, Youbang Sun, Hongyi Liu, Yuntian Wei, Zhekai Chen, Xuekai Zhu, Kaiyan Zhang, Bingning Wang, Ning Ding, and Bowen Zhou. 2025. Towards a unified view of large language model post-training. *Preprint*, arXiv:2509.04419.

Hongbo Ma, Bangji Yang, Yunqian Selina Cheng, Jiajun Fan, Hanwen Zhang, and Ge Liu. 2026. Constraint-first reasoning: A training-free protocol for exploiting answer-space constraints in mathematical problem solving. *Preprint*, arXiv:2608.05254.

Lu Ma, Hao Liang, Meiyi Qiang, Lexiang Tang, Xiaochen Ma, Zhen Hao Wong, Junbo Niu, Chengyu Shen, Runming He, Yanhao Li, Bin Cui, and Wentao Zhang. 2025a. Learning what reinforcement learning can't: Interleaved online fine-tuning for hardest questions. *Preprint*, arXiv:2506.07527.

Xueguang Ma, Qian Liu, Dongfu Jiang, Ge Zhang, Zejun Ma, and Wenhu Chen. 2025b. General-Reasoner: Advancing LLM reasoning across all domains. *Preprint*, arXiv:2505.14652.

Pulkit Madan, Sanjay Haresh, Reza Ebrahimi, Sunny Panchal, Apratim Bhattacharyya, and Roland Memisevic. 2026. On locality and length generalization in visual reasoning. *Preprint*, arXiv:2607.09061.

Eran Malach, Omid Saremi, Sinead Williamson, Arwen Bradley, Aryo Lotfi, Emmanuel Abbe, Josh Susskind, and Etai Littwin. 2026. To infinity and beyond: Tool-use unlocks length generalization in state space models. In *International Conference on Learning Representations*.

Mayug Maniparambil, Arjun Karuvally, Terrence Sejnowski, and Fergal Reid. 2026. When RL suppresses its own vocabulary: Recovering reasoning diversity in puzzle-to-math transfer. *Preprint*, arXiv:2605.29190.

Kohsei Matsutani, Gouki Minegishi, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo. 2026. Zipping the thought: When and how compressed reasoning data works in LLM post-training. *Preprint*, arXiv:2605.28008.

R. Thomas McCoy, Shunyu Yao, Dan Friedman, Mathew D. Hardy, and Thomas L. Griffiths. 2024. Embers of autoregression show how large language models are shaped by the problem they are trained to solve. *Proceedings of the National Academy of Sciences*.

Sean McLeish, Arpit Bansal, Alex Stein, Neel Jain, John Kirchenbauer, Brian R. Bartoldson, Bhavya Kailkhura, Abhinav Bhatele, Jonas Geiping, Avi Schwarzschild, and Tom Goldstein. 2024. Transformers can do arithmetic with the right embeddings. In *Advances in Neural Information Processing Systems*, volume 37, pages 108012–108041.

Manas Mehta, Fangcong Yin, and Greg Durrett. 2026. Randomized YaRN improves length generalization for long-context reasoning. *Preprint*, arXiv:2606.23687.

William Merrill and Ashish Sabharwal. 2024. The expressive power of transformers with chain of thought. In *International Conference on Learning Representations*.

Anna Mészáros, Szilvia Ujváry, Wieland Brendel, Patrik Reizinger, and Ferenc Huszár. 2024. Rule extrapolation in language models: A study of compositional generalization on OOD prompts. *Preprint*, arXiv:2409.13728.

Elliot Meyerson, Giuseppe Paolo, Roberto Dailey, Hormoz Shahrzad, Olivier Francon, Conor F. Hayes, Xin Qiu, Babak Hodjat, and Risto Miikkulainen. 2025. Solving a million-step LLM task with zero errors. *Preprint*, arXiv:2511.09030.

Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. 2025. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models. In *International Conference on Learning Representations*.

Mehryar Mohri. 2026. Rational transductors. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 89841–89876. PMLR.

Philipp Mondorf, Shijia Zhou, Monica Riedler, and Barbara Plank. 2025. Compositional-ARC: Assessing systematic generalization in abstract spatial reasoning. *Preprint*, arXiv:2504.01445.

Mohammad Mahdi Moradi, Hossam Amer, Sudhir Mudur, Weiwei Zhang, Yang Liu, and Walid Ahmed. 2025. Continuous self-improvement of large language models by test-time training with verifier-driven sample selection. *Preprint*, arXiv:2505.19475.

Maximilian Mordig, Andreas Opedal, Weiyang Liu, and Bernhard Schölkopf. 2026. Rethinking easy-to-hard: Limits of curriculum learning in post-training for deductive reasoning. *Preprint*, arXiv:2603.27226.

Terufumi Morishita, Gaku Morio, Atsuki Yamaguchi, and Yasuhiro Sogawa. 2024. Enhancing reasoning capabilities of LLMs via principled synthetic logic corpus. In *Advances in Neural Information Processing Systems*, volume 37.

Sumeet Ramesh Motwani, Alesia Ivanova, Ziyang Cai, Philip Torr, Riashat Islam, Shital Shah, Christian Schroeder de Witt, and Charles London. 2026. h1: Bootstrapping LLMs to reason over longer horizons via reinforcement learning. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*. PMLR.

Sajad Movahedi, Vera Milovanović, Shlomo Libo Feigin, Alexander Theus, Thomas Hofmann, Valentina Boeva, T. Konstantin Rusch, and Antonio Orvieto. 2026. Fixed-point reasoners: Stable and adaptive deep looped transformers. *Preprint*, arXiv:2606.18206.

Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. 2025. s1: Simple test-time scaling. *Preprint*, arXiv:2501.19393.

Dor Muhlgay, Ori Ram, Inbal Magar, Yoav Levine, Nir Ratner, Yonatan Belinkov, Omri Abend, Kevin Leyton-Brown, Amnon Shashua, and Yoav Shoham. 2024. Generating benchmarks for factuality evaluation of language models. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 49–66. Association for Computational Linguistics.

Shikhar Murty, Pratyusha Sharma, Jacob Andreas, and Christopher D. Manning. 2023. Grokking of hierarchical structure in vanilla transformers. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 439–448.

Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. 2023. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*.

Yannic Neuhaus, Nicolas Flammarion, Matthias Hein, and Francesco Croce. 2026. On the out-of-distribution generalization of reasoning in multimodal LLMs for simple visual planning tasks. *Preprint*, arXiv:2602.15460.

Kangqi Ni, Zhen Tan, Zijie Liu, Pingzhi Li, and Tianlong Chen. 2025. Can GRPO help LLMs transcend their pretraining origin? *Preprint*, arXiv:2510.15990.

Alexandru Oarga and Yilun Du. 2025. Generalizable reasoning through compositional energy minimization. *Preprint*, arXiv:2510.20607.

Andreas Opedal, Haruki Shirakami, Bernhard Schölkopf, Abulhair Saparov, and Mrinmaya Sachan. 2025. MathGAP: Out-of-distribution evaluation on problems with arbitrarily complex proofs. In *International Conference on Learning Representations*.

Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long T. Le, Samira Daruki, Xiangru Tang, Vishy Tirumalashetty, George Lee, Mahsan Rofouei, Hangfei Lin, Jiawei Han, Chen-Yu Lee, and Tomas Pfister. 2026. ReasoningBank: Scaling agent self-evolving with reasoning memory. In *International Conference on Learning Representations*.

Simon Park, Simran Kaur, and Sanjeev Arora. 2025. How does RL post-training induce skill composition? A case study on Countdown. *Preprint*, arXiv:2512.01775.

Miao Peng, Nuo Chen, Zongrui Suo, and Jia Li. 2025. Rewarding graph reasoning process makes LLMs more generalized reasoners. *Preprint*, arXiv:2503.00845.

Ruiying Peng, Mengyu Yang, Jing Lei, Xiaohui Li, Xueyu Wu, and Xinlei Chen. 2026. Memorize theorems, not instances: Probing SFT generalization through mathematical reasoning. *Preprint*, arXiv:2605.09270.

Long Phan et al. 2026. A benchmark of expert-level academic questions to assess AI capabilities. *Nature*, 649(8099):1139–1146.

Leonard S. Pleiss, Maximilian Schiffer, and Robert K. von Weizsaecker. 2026. Disentangling generalization and memorization in large language models using chess. *Preprint*, arXiv:2601.16823.

Andrei Cristian Popescu, Haitz Sáez de Ocáriz Borde, and Pietro Liò. 2026. Looped language models improve compositional tool calling. *Preprint*, arXiv:2608.18171.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. 2023. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711.

Cheng Qian, Emre Can Acikgoz, Hongru Wang, Xiusi Chen, Avirup Sil, Dilek Hakkani-Tür, Gokhan Tur, and Heng Ji. 2025. Smart: Self-aware agent for tool overuse mitigation. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 4604–4621.

Nived Rajaraman, Audrey Huang, Miroslav Dudik, Robert Schapire, Dylan Foster, and Akshay Krishnamurthy. 2026. Learning to reason with curriculum II: Compositional generalization. *Preprint*, arXiv:2606.27721.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. 2024. GPQA: A graduate-level Google-proof Q&A benchmark. In *First Conference on Language Modeling*.

Aswin RRV, Jacob Dineen, Divij Handa, Mihir Parmar, Ben Zhou, Swaroop Mishra, and Chitta Baral. 2026. Mid-training with self-generated data improves reinforcement learning in language models. *Preprint*, arXiv:2605.08472.

Mahdi Sabbaghi, George Pappas, Hamed Hassani, and Surbhi Goel. 2024. Explicitly encoding structural symmetry is key to length generalization in arithmetic tasks. *Preprint*, arXiv:2406.01895.

Moein Salimi, Danial Parnian, Shaygan Adim, Amirmohammad Ebrahiminasab, Nima Alighardashi, Parsa Gholami, Sahand Akramipour, Mahdi Jafari Siavoshani, and Mohammad Hossein Rohban. 2026. CEDAR-GRPO: Process-aware reinforcement learning for general abductive reasoning in LLMs. *Preprint*, arXiv:2608.14791.

Abulhair Saparov and He He. 2023. Testing the general deductive reasoning capacity of large language models using OOD examples. In *Advances in Neural Information Processing Systems*, volume 36.

Kusha Sareen, Morgane M. Moss, Alessandro Sordoni, Rishabh Agarwal, and Arian Hosseini. 2025. Putting the value back in RL: Better test-time scaling by unifying LLM reasoners with verifiers. *Preprint*, arXiv:2505.04842.

Yash Sarrof, Yupei Du, Katharina Stein, Alexander Koller, Sylvie Thiébaux, and Michael Hahn. 2026. On the ability of transformers to verify plans. *Preprint*, arXiv:2603.19954.

Dustin Schwenk, Apoorv Khandelwal, Christopher Clark, Kenneth Marino, and Roozbeh Mottaghi. 2022. A-OKVQA: A benchmark for visual question answering using world knowledge. In *Computer Vision – ECCV 2022*, pages 146–162. Springer.

Mahnoor Shahid and Hannes Rothe. 2026. AGELComp: A neuro-symbolic framework for compositional generalization in interactive agents. *Preprint*, arXiv:2604.26522.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *Preprint*, arXiv:2402.03300.

Mark Shapiro. 2026. Retrofitting recurrent depth into a pretrained language model: Installation, extrapolation, transfer, and retention at two parameter budgets. *Preprint*, arXiv:2608.11233.

Aman Sharma and Paras Chopra. 2026. EsoLangBench: Evaluating genuine reasoning in large language models via esoteric programming languages. *Preprint*, arXiv:2603.09678.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*.

Shlok Shelat, Jay Raval, Souvik Roy, and Manas Gaur. 2026. Beyond memorization: Testing LLM reasoning on unseen theory of computation tasks. *Preprint*, arXiv:2601.13392.

Ruoqi Shen, Sébastien Bubeck, Ronen Eldan, Yin Tat Lee, Yuanzhi Li, and Yi Zhang. 2023. Positional description matters for transformers arithmetic. *Preprint*, arXiv:2311.14737.

Dan Shi, Zhuowen Han, Simon Ostermann, Renren Jin, Josef van Genabith, and Deyi Xiong. 2026. Why does reinforcement learning generalize? A feature-level mechanistic study of post-training in large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 38979–39000.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*. PMLR.

Atsushi Shimizu, Shohei Taniguchi, and Yutaka Matsuo. 2026. Position encoding with random float sampling enhances length generalization of transformers. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 4966–4980, Rabat, Morocco. Association for Computational Linguistics.

Parshin Shojaee, Iman Mirzadeh, Keivan Alizadeh, Maxwell Horton, Samy Bengio, and Mehrdad Farajtabar. 2025. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity. In *Advances in Neural Information Processing Systems*, volume 38.

Safal Shrestha, Minwu Kim, Aadim Nepal, Anubhav Shrestha, and Keith W. Ross. 2025. Warm up before you train: Unlocking general reasoning in resource-constrained settings. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 14380–14401.

Subham Singh, Ashutosh Mishra, and Subha Raut. 2026. Relative positions generalize, absolute positions memorize: An implicit-bias account of length generalization in attention. *Preprint*, arXiv:2607.18759.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. 2024. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. *Preprint*, arXiv:2408.03314.

Kaya Stechly, Karthik Valmeekam, and Subbarao Kambhampati. 2024. Chain of thoughtlessness? An analysis of CoT in planning. In *Advances in Neural Information Processing Systems*, volume 37.

Julius Steiglechner, Lucas Mahler, and Gabriele Lohmann. 2026. LLMs can see the smoke but not the fire: Evaluating abductive reasoning with Elenchos. *Preprint*, arXiv:2607.12733.

Zafir Stojanovski, Oliver Stanley, Joe Sharratt, Richard Jones, Abdulhakeem Adefioye, Jean Kaddour, and Andreas Köpf. 2025. Reasoning Gym: Reasoning environments for reinforcement learning with verifiable rewards. *Preprint*, arXiv:2505.24760.

Yiyou Sun, Shawn Hu, Georgia Zhou, Ken Zheng, Hannaneh Hajishirzi, Nouha Dziri, and Dawn Song. 2025. OMEGA: Can LLMs reason outside the box in math? Evaluating exploratory, compositional, and transformative generalization. *Preprint*, arXiv:2506.18880.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. 2023. Challenging BIG-Bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13003–13051.

Shota Takashiro, Masanori Koyama, Takeru Miyato, Yusuke Iwasawa, Yutaka Matsuo, and Kohei Hayashi. 2026. Exploration of fast-slow latent recurrence for train-short, test-long generalization. *Preprint*, arXiv:2604.01577.

Xueyun Tian, Minghua Ma, Bingbing Xu, Nuoyan Lyu, Wei Li, Heng Dong, Zheng Chu, Yuanzhuo Wang, and Huawei Shen. 2026. Learning from mistakes: Negative reasoning samples enhance out-of-domain generalization. *Preprint*, arXiv:2601.04992.

Ruilin Tong and Dong Gong. 2026. MILES: Modular instruction memory with learnable selection for self-improving LLM reasoning. *Preprint*, arXiv:2607.06974.

Yao Tong, Jiayuan Ye, Anastasia Borovykh, and Reza Shokri. 2026. Generalization in LLM problem solving: The case of the shortest path. *Preprint*, arXiv:2604.15306.

Eduardo Valle and Fergal Reid. 2026. Protoreasoning in tiny transformers. *Preprint*, arXiv:2608.04980.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30.

Petar Veličković, Adrià Puigdomènech Badia, David Budden, Razvan Pascanu, Andrea Banino, Misha Dashevskiy, Raia Hadsell, and Charles Blundell. 2022. The CLRS algorithmic reasoning benchmark. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*. PMLR.

Ivan Viakhirev, Kirill Borodin, Amirah Almutairi, Serguei Barannikov, Maxim Abramov, and Grach Mkrtchian. 2026. Think shallow, solve deep: Controlling recurrent dynamics for reliable test-time depth. *Preprint*, arXiv:2608.18222.

Boshi Wang, Xiang Yue, Yu Su, and Huan Sun. 2024a. Grokking of implicit reasoning in transformers: A mechanistic journey to the edge of generalization. In *Advances in Neural Information Processing Systems*, volume 37.

Peng-Yuan Wang, Tian-Shuo Liu, Chenyang Wang, Yi-Di Wang, Shu Yan, Cheng-Xing Jia, Xu-Hui Liu, Xin-Wei Chen, Jia-Cheng Xu, Ziniu Li, and Yang Yu. 2025. A survey on large language models for mathematical reasoning. *Preprint*, arXiv:2506.08446.

Ru Wang, Wei Huang, Selena Song, Haoyu Zhang, Qian Niu, Yusuke Iwasawa, Yutaka Matsuo, and Jiaxian Guo. 2026a. Beyond in-distribution success: Scaling curves of CoT granularity for language model generalization. In *Proceedings of the Third Conference on Parsimony and Learning*, volume 328 of *Proceedings of Machine Learning Research*, pages 582–611. PMLR.

Rui Wang, Hongru Wang, Boyang Xue, Yixia Li, Jianhui Pang, Shudong Liu, Yi Chen, Jiahao Qiu, Derek Fai Wong, Guanhua Chen, Heng Ji, and Kam-Fai Wong. 2026b. Harnessing the reasoning economy: A survey of efficient reasoning for large language models. *Preprint*, arXiv:2503.24377.

Shaojie Wang and Liang Zhang. 2026. From meta-thought to execution: Cognitively aligned post-training for generalizable and reliable LLM reasoning. *Preprint*, arXiv:2601.21909.

Tianle Wang, Zhaoyang Wang, Guangchen Lan, Xinpeng Wei, Sipeng Zhang, Guanwen Qiu, and Abulhair Saparov. 2026c. How deep can LLMs learn to reason? Expressiveness is key. *Preprint*, arXiv:2605.06638.

Xinye Wang, Junxiao Liu, and Shujian Huang. 2026d. RP-OPSD: Reasoning-pivot-guided on-policy self-distillation for multilingual reasoning transfer. *Preprint*, arXiv:2608.06347.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyan Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhu Chen. 2024b. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems*, volume 37.

Yifan Wei, Li Du, Xiaoyan Yu, Yang Feng, and Angsheng Li. 2026. Towards compositional generalization of LLMs via skill taxonomy guided data synthesis. *Preprint*, arXiv:2601.03676.

Johannes Welbl, Nelson F. Liu, and Matt Gardner. 2017. Crowdsourcing multiple choice science questions. In *Proceedings of the 3rd Workshop on Noisy User-generated Text*, pages 94–106.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Ben Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, Shubh-Agrawal, Sandeep Singh Sandha, Siddartha Naidu, Chinmay Hegde, Yann LeCun, Tom Goldstein, Willie Neiswanger, and Micah Goldblum. 2024. LiveBench: A challenging, contamination-limited LLM benchmark. *Preprint*, arXiv:2406.19314.

Menghua Wu, Cai Zhou, Stephen Bates, and Tommi Jaakkola. 2025. Thought calibration: Efficient and confident test-time scaling. *Preprint*, arXiv:2505.18404.

Xinyi Wu, Siyuan Liu, and Ali Jadbabaie. 2026. How data shapes RoPE frequency usage: From positional scale matching to length generalization. *Preprint*, arXiv:2607.07678.

Zhaofeng Wu, Linlu Qiu, Alexis Ross, Ekin Akyürek, Boyuan Chen, Bailin Wang, Najoung Kim, Jacob Andreas, and Yoon Kim. 2024. Reasoning or reciting? Exploring the capabilities and limitations of language models through counterfactual tasks. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1819–1862.

Hao Xiang, Qiaoyu Tang, Le Yu, Yaojie Lu, Xianpei Han, Ben He, Le Sun, Bowen Yu, Peng Wang, Hongyu Lin, and Dayiheng Liu. 2026. Verifiable environments are LEGO bricks: Recursive composition for reasoning generalization. *Preprint*, arXiv:2606.12373.

Tian Xie, Zitian Gao, Qingnan Ren, Haoming Luo, Yuqian Hong, Bryan Dai, Joey Zhou, Kai Qiu, Zhirong Wu, and Chong Luo. 2025. Logic-RL: Unleashing LLM reasoning with rule-based reinforcement learning. *Preprint*, arXiv:2502.14768.

Ruize Xu, Chenxiao Yang, Yanhong Li, and David McAllester. 2026a. Training transformers as a universal computer. *Preprint*, arXiv:2604.25166.

Weiwen Xu, Jia Liu, Hou Pong Chan, Long Li, Deng Cai, Min Chen, and Hao Zhang. 2026b. Beyond entropy: Correctness-aware advantage shaping via contrastive policy optimization. *Preprint*, arXiv:2607.14614.

Yongzhong Xu. 2026. Early-warning signals of grokking via loss-landscape geometry. *Preprint*, arXiv:2602.16967.

Cheng Yan, Zhijun Fan, Guangyang Ye, Fan Xu, Xiang Xia, Yawei Wang, and Wuyang Zhang. 2026. UPAIR: Diagnosing reasoning states via uncertainty-progress alignment for selective intervention. *Preprint*, arXiv:2607.17188.

Jianhao Yan, Yafu Li, Zican Hu, Zhi Wang, Ganqu Cui, Xiaoye Qu, Yu Cheng, and Yue Zhang. 2025. Learning to reason under off-policy guidance. *Preprint*, arXiv:2504.14945.

An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu, Xingzhang Ren, and Zhenru Zhang. 2024. Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement. *Preprint*, arXiv:2409.12122.

Andy Yang, Pascal Bergsträßer, Georg Zetzsche, David Chiang, and Anthony W. Lin. 2026a. Length generalization bounds for transformers. *Preprint*, arXiv:2603.02238.

Andy Yang, Blerta Veseli, Corentin Barloy, Michaël Cadilhac, Andreas Krebs, Charles Paperman, Howard Straubing, and Michael Hahn. 2026b. Algebraic decomposition theory for transformer length generalization. *Preprint*, arXiv:2608.13433.

Xiao-Wen Yang, Jie-Jing Shao, Lan-Zhe Guo, Bo-Wen Zhang, Zhi Zhou, Lin-Han Jia, Wang-Zhou Dai, and Yu-Feng Li. 2025. Neuro-symbolic artificial intelligence: Towards improving the reasoning abilities of large language models. *Preprint*, arXiv:2508.13678.

Xinhao Yao, Ruifeng Ren, Yun Liao, Lizhong Ding, and Yong Liu. 2025a. Compositional generalization from learned skills via CoT training: A theoretical and structural analysis for reasoning. *Preprint*, arXiv:2502.04667.

Yihang Yao, Zhepeng Cen, Miao Li, William Han, Yuyou Zhang, Emerson Liu, Zuxin Liu, Chuang Gan, and Ding Zhao. 2025b. Your language model may think too rigidly: Achieving reasoning consistency with symmetry-enhanced training. In *Findings of the Association for Computational Linguistics: ACL 2025*.

Tian Ye, Zicheng Xu, Yuanzhi Li, and Zeyuan Allen-Zhu. 2025a. Physics of language models: Part 2.1, grade-school math and the hidden reasoning process. In *International Conference on Learning Representations*.

Yixin Ye, Zhen Huang, Yang Xiao, Ethan Chern, Shijie Xia, and Pengfei Liu. 2025b. LIMO: Less is more for reasoning. *Preprint*, arXiv:2502.03387.

Fangcong Yin, Zeyu Leo Liu, Liu Leqi, Xi Ye, and Greg Durrett. 2025. Learning composable chains-of-thought. *Preprint*, arXiv:2505.22635.

Qifan Yu, Zhenyu He, Sijie Li, Zhou Xun, Jun Zhang, Jingjing Xu, and Di He. 2026. Enhancing autoregressive chain-of-thought through loop-aligned reasoning. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2206–2222, Rabat, Morocco. Association for Computational Linguistics.

Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, Haibin Lin, Zhiqi Lin, Bole Ma, Guangming Sheng, Yuxuan Tong, Chi Zhang, Mofan Zhang, Wang Zhang, and 16 others. 2025. DAPO: An open-source LLM reinforcement learning system at scale. *Preprint*, arXiv:2503.14476.

Lifan Yuan, Weize Chen, Yuchen Zhang, Ganqu Cui, Hanbin Wang, Ziming You, Ning Ding, Zhiyuan Liu, Maosong Sun, and Hao Peng. 2025. From $f(x)$ and $g(x)$ to $f(g(x))$: LLMs learn new skills in RL by composing old ones. *Preprint*, arXiv:2509.25123.

Zike Yuan, Han Zhang, Jianzhi Yan, Le Liu, Cai Ke, Huozhi Zhou, Jian Xie, Jiran Yin, Yukun Cao, Yue Yu, Hui Wang, Ming Liu, and Bing Qin. 2026. GRAIN: Bridging name and narrative shifts in real-world graph reasoning through invariance-rewarded agentic RL. *Preprint*, arXiv:2608.27142.

Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. 2025. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? *Preprint*, arXiv:2504.13837.

Boyi Zeng, Shixiang Song, Siyuan Huang, Yixuan Wang, He Li, Ziwei He, Xinbing Wang, Zhiyu Li, and Zhouhan Lin. 2026. PonderLM: Pretraining language models to ponder in continuous space. In *International Conference on Learning Representations*.

Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, and Junxian He. 2025. SimpleRL-Zoo: Investigating and taming zero reinforcement learning for open base models in the wild. *Preprint*, arXiv:2503.18892.

Georg Zetzsche, Hongjian Jiang, Andy Yang, Pascal Bergsträßer, Marco Sälzer, David Chiang, and Anthony W. Lin. 2026. Length generalization for transformers via compression. *Preprint*, arXiv:2609.08851.

Kaiwen Zha, Zhengqi Gao, Maohao Shen, Zhang-Wei Hong, Duane S. Boning, and Dina Katabi. 2025. RL Tango: Reinforcing generator and verifier together for language reasoning. *Preprint*, arXiv:2505.15034.

Charlie Zhang, Graham Neubig, and Xiang Yue. 2025a. On the interplay of pre-training, mid-training, and RL on reasoning language models. *Preprint*, arXiv:2512.07783.

Dan Zhang, Sining Zhoubian, Ziniu Hu, Yisong Yue, Yuxiao Dong, and Jie Tang. 2024a. ReST-MCTS*: LLM self-training via process reward guided tree search. In *Advances in Neural Information Processing Systems*, volume 37.

Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson, Catherine Wu, Will Song, Tiffany Zhao, Pranav Raja, Charlotte Zhuang, Dylan Slack, Qin Lyu, Sean Hendryx, Russell Kaplan, Michele Lunati, and Summer Yue. 2024b. A careful examination of large language model performance on grade school arithmetic. *Preprint*, arXiv:2405.00332.

Kaiyi Zhang, Wei Wu, and Yankai Lin. 2026a. DelTA: Discriminative token credit assignment for reinforcement learning from verifiable rewards. *Preprint*, arXiv:2605.21467.

Minxing Zhang, Yi Yang, Roy Xie, Bhuwan Dhingra, Shuyan Zhou, and Jian Pei. 2025b. Generalizability of large language model-based agents: A comprehensive survey. *Preprint*, arXiv:2509.16330.

Yifan Zhang, Wenyu Du, Dongming Jin, Jie Fu, and Zhi Jin. 2025c. Finite state automata inside transformers with chain-of-thought: A mechanistic study on state tracking. *Preprint*, arXiv:2502.20129.

Yizhou Zhang, Lun Du, Defu Cao, Qiang Fu, and Yan Liu. 2024c. An examination on the effectiveness of divide-and-conquer prompting in large language models. *Preprint*, arXiv:2402.05359.

Zhongwang Zhang, Pengxiao Lin, Zhiwei Wang, Yaoyu Zhang, and Zhi-Qin John Xu. 2024d. Initialization is critical to whether transformers fit composite functions by reasoning or memorizing. *Preprint*, arXiv:2405.05409.

Zhongwang Zhang, Pengxiao Lin, Zhiwei Wang, Yaoyu Zhang, and Zhi-Qin John Xu. 2025d. Complexity control facilitates reasoning-based compositional generalization in transformers. *Preprint*, arXiv:2501.08537.

Zili Zhang, Yilin Wang, Heng Wang, Herun Wan, and Minnan Luo. 2026b. Why knowing both hops is not enough: Understanding two-hop generalization in language models. *Preprint*, arXiv:2608.07261.

Chengshuai Zhao, Zhen Tan, Pingchuan Ma, Dawei Li, Bohan Jiang, Yancheng Wang, Yingzhen Yang, and Huan Liu. 2025. Is chain-of-thought reasoning of LLMs a mirage? A data distribution lens. *Preprint*, arXiv:2508.01191.

Guangxiang Zhao, Qilong Shi, Xusen Xiao, Wenpu Liu, Yaoming Li, Linfeng Hao, Shuyang Hou, Zijian Guo, Xinrui Zhang, Yuntian Zhao, Zhengyang Wang, Wenrui Liu, Yuhan Wu, Tong Yang, Lin Sun, and Xiangzheng Zhang. 2026a. ScienceArena: Benchmarking LLMs on latest scientific olympiad competitions. *Preprint*, arXiv:2608.30517.

Jinze Zhao, Peihao Wang, Junjie Yang, Ruisi Cai, Gaowen Liu, Jayanth Srinivasa, Ramana Rao Kompella, Yingbin Liang, and Zhangyang Wang. 2026b. Sparse mixture-of-experts for compositional generalization: Empirical evidence and theoretical foundations of optimal sparsity. In *Proceedings of the Third Conference on Parsimony and Learning*, volume 328 of *Proceedings of Machine Learning Research*, pages 1049–1071. PMLR.

Ziqi Zhao, Xinyu Ma, Liu Yang, Yujie Feng, Daiting Shi, Jingzhou He, Xin Xin, Zhaochun Ren, and Xiao-Ming Wu. 2026c. Beyond imitation: Reflective on-policy self-distillation for LLM reasoning. *Preprint*, arXiv:2605.28014.

Huaixiu Steven Zheng, Swaroop Mishra, Hugh Zhang, Xinyun Chen, Minmin Chen, Azade Nova, Le Hou, Heng-Tze Cheng, Quoc V. Le, Ed H. Chi, and Denny Zhou. 2024. Natural Plan: Benchmarking LLMs on natural language planning. *Preprint*, arXiv:2406.04520.

Cai Zhou, Zekai Wang, Menghua Wu, Qianyu Julie Zhu, Flora C. Shi, Chenyu Wang, Ashia Wilson, Tommi Jaakkola, and Stephen Bates. 2026. Online reasoning calibration: Test-time training enables generalizable conformal LLM reasoning. *Preprint*, arXiv:2604.01170.

Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc Le, and Ed Chi. 2023. Least-to-most prompting enables complex reasoning in large language models. In *International Conference on Learning Representations*.

Hattie Zhou, Arwen Bradley, Etai Littwin, Noam Razin, Omid Saremi, Josh Susskind, Samy Bengio, and Preetum Nakkiran. 2024a. What algorithms can transformers learn? A study in length generalization. In *International Conference on Learning Representations*.

Pei Zhou, Jay Pujara, Xiang Ren, Xinyun Chen, Heng-Tze Cheng, Quoc V. Le, Ed H. Chi, Denny Zhou, Swaroop Mishra, and Huaixiu Steven Zheng. 2024b. Self-Discover: Large language models self-compose reasoning structures. In *Advances in Neural Information Processing Systems*, volume 37, pages 126032–126058.

Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024c. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations*.

Yongchao Zhou, Uri Alon, Xinyun Chen, Xuezhi Wang, Rishabh Agarwal, and Denny Zhou. 2024d. Transformers can achieve length generalization but not robustly. *Preprint*, arXiv:2402.09371.

Yujun Zhou, Zhenwen Liang, Haolin Liu, Wenhao Yu, Kishan Panaganti, Linfeng Song, Dian Yu, Xiangliang Zhang, Haitao Mi, and Dong Yu. 2025. Evolving language models without labels: Majority drives selection, novelty promotes variation. *Preprint*, arXiv:2509.15194.

Jiachen Zhu, Congmin Zheng, Jianghao Lin, Kounianhua Du, Ying Wen, Yong Yu, Jun Wang, and Weinan Zhang. 2025a. Retrieval-augmented process reward model for generalizable mathematical reasoning. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 8453–8468, Vienna, Austria. Association for Computational Linguistics.

Jiajun Zhu, Peihao Wang, Ruisi Cai, Jason D. Lee, Pan Li, and Zhangyang Wang. 2025b. Rethinking addressing in language models via contextualized equivariant positional encoding. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 80155–80186. PMLR.

Rui-Jie Zhu, Tianhao Peng, Tianhao Cheng, Xingwei Qu, Jinfa Huang, Dawei Zhu, Hao Wang, Kaiwen Xue, Xuanliang Zhang, Yong Shan, Tianle Cai, Taylor Kergan, Assel Kembay, Andrew Smith, Chenghua Lin, Binh Nguyen, Yuqi Pan, Yuhong Chou, Zefan Cai, and 14 others. 2025c. A survey on latent reasoning. *Preprint*, arXiv:2507.06203.

Taojie Zhu, Dongyang Xu, Ding Zou, Sen Zhao, Qiaobo Hao, Zhiguo Yang, and Yonghong He. 2026. Bridging SFT and RL: Dynamic policy optimization for robust reasoning. *Preprint*, arXiv:2604.08926.

A Literature Collection and Corpus

A.1 Abstract-Based Retrieval

We searched arXiv abstracts using terms such as "reasoning generalization", "reasoning transfer", "OOD reasoning", "compositional generalization", "systematic generalization", and "length generalization". Supplementary searches used "post-training", "SFT", "GRPO", "RLVR", and "reward" to broaden coverage of training methods.

A.2 Screening and Corpus Construction

After deduplication, we screened abstracts and read relevant papers to assess their methods and evidence of reasoning transfer. The *corpus version* contains 239 studies across the four pillars, including contextual methods without direct transfer evidence. We selected 212 studies for the *survey version*, covering the main text and appendix evidence. Each paper is counted once under its primary pillar; background, survey, and dataset-only references are excluded. Figure 4 and Table 1 summarize both versions.

Pillar	Version	≤2023	2024	2025	2026	Total
Training	Corpus	0	5	42	51	98
	Survey	0	4	34	38	76
Inference	Corpus	2	7	9	7	25
	Survey	2	7	9	7	25
Architecture	Corpus	6	7	5	22	40
	Survey	6	7	5	21	39
Analysis	Corpus	13	16	15	32	76
	Survey	13	16	15	28	72
Total	Corpus	21	35	71	112	239
	Survey	21	34	63	94	212

Table 1: Catalogue and survey studies by primary pillar and first-posting year. Venue year is used for records without an arXiv version. Background references and related surveys are excluded from both versions.

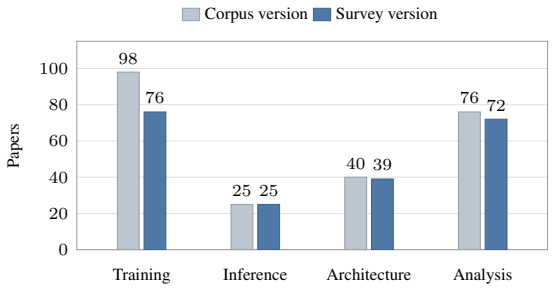


Figure 4: The corpus version ($n = 239$) and author-selected survey version ($n = 212$), by primary pillar. Survey counts include main-text and appendix studies. Papers cited across pillars are counted once.

Figure 1 shows representative works, including contextual methods. Table 2 lists selected catalogue studies covered outside the main discussion, grouped by their primary branch.

B Synthesis and Related Surveys

B.1 Connections Across Pillars

Training and architecture determine which procedures can be reused at inference. Inductive scratchpads improve length transfer by making each step depend only on the question and previous state (Abbe et al., 2024), while recurrent computation requires stable updates as depth increases (Viakhirev

Branch	Catalogue studies
Training (§3) and inference (§4)	
Mid-training	Mehta et al. (2026)
Reinforcement learning	Yuan et al. (2025), Li et al. (2026a), Chaudhry et al. (2026)
Trajectory restructuring	Hou et al. (2026)
Architecture (§5)	
Recurrent depth: weight sharing	Chen (2026), Yu et al. (2026)
Recurrent depth: adaptive computation and stability	Abnar et al. (2023), Kuo et al. (2026), Ibrahimli et al. (2026), Huang et al. (2026a), Labovich (2026), Movahedi et al. (2026)
Information routing: positional encoding and locality	Wu et al. (2026), Shimizu et al. (2026), Madan et al. (2026)
Information routing: modular and neuro-symbolic	Kamali et al. (2025), Shahid and Rothe (2026), Oarga and Du (2025)
Recurrent memory	Xu et al. (2026a), Takashiro et al. (2026)
Analysis (§6)	
Compositional generalization	Saparov and He (2023), Fu and Frank (2024b)
Length, depth, and difficulty	Ding et al. (2024), Stechly et al. (2024), Zheng et al. (2024), Shojaee et al. (2025), Ye et al. (2025a)
Prior exposure	McCoy et al. (2024), Chi et al. (2024), Shelat et al. (2026)
Shared circuits	Zhang et al. (2025c), Yao et al. (2025a)
Learning dynamics	Nanda et al. (2023), Xu (2026), Shi et al. (2026)
Learnability and certification	Guan and Bradic (2026), Zetzsche et al. (2026)

Table 2: Additional catalogue studies cited in the survey, grouped by their primary branch in Figure 1.

et al., 2026). External memory can extend the length range of bounded-state models, but this establishes a capability of the combined system rather than the unaided model (Malach et al., 2026).

Training must also support the inference procedure used at deployment. In controlled shortest-path tasks, test-time scaling remains limited by the reasoning learned during training (Tong et al., 2026). Controller-aware training addresses this mismatch and improves transfer to new inference workflows (Choudhury et al., 2026). Tables 3 and 4 summarize the transfer mechanisms, limitations, and evaluation controls of the training and inference branches.

B.2 Comparison with Related Surveys

Table 5 compares representative surveys of reasoning tasks and methods. Their coverage overlaps with ours, but our emphasis is the transfer of reasoning procedures under explicit changes in composition, length, domain, or problem conventions. We connect the methods reviewed with behavioral, mechanistic, and theoretical evidence about the conditions under which they generalize.

Training branch	Transfer mechanism	Strongest current evidence	Main risk
Pre-training	Establish reusable capabilities before adaptation	Controlled training-stage and large-pass@k comparisons	Opaque exposure makes the boundary hard to identify
Mid-training and curricula	Expose positions, procedures, or domains absent from base training	Long-context and verified-synthesis transfer	More data may repeat the same shortcut
SFT and distillation	Teach explicit operations and reachable traces	Checkpoint sweeps, cross-model transfer, logically isomorphic OOD tests	Teacher forcing, late-stage overfitting, suppressed uncertainty
RL and RLVR	Select and sometimes compose procedures with outcome or process feedback	Held-out domains, capability-controlled pass@k, composed environments	Reward hacking, entropy collapse, domain interference, dependence on base competence
Hybrid SFT/RL	Combine demonstrations with on-policy exploration	Shared ID/OOD protocol on OpenR1-Math-46k (Table 10)	Gains shrink against correctly implemented SFT→GRPO; single-run reproductions
Self-improvement and controller-aware training	Learn reusable skills or cooperation with inference controllers	Unseen skill combinations and held-out controller compositions	Specialization to the combinations seen in training

Table 3: Training branches: how each is expected to support transfer, the strongest current evidence, and the main risk.

Inference branch	Helps when	Fails when	Required control
Trajectory restructuring	The task representation is misaligned or intermediate structure is missing	The decomposition is wrong or prompt-sensitive	Matched prompts and paraphrase tests
Compute scaling	Correct paths exist with low but nonzero probability	No useful candidate is generated	pass@k and an oracle-selection upper bound
State assessment	Errors are easier to detect than to avoid, or difficulty varies across examples	The verifier shares the generator's shortcut, or the monitor is specific to the source distribution	Independent checks and source-to-target accuracy-cost frontiers
Externalization	Exact operations or reusable skills can be moved outside the model	Tool routing or retrieval fails	Tool-free and retrieval-distance controls

Table 4: Inference branches (§4): when each helps, when it fails, and the control needed to claim transfer.

Survey	Main focus
Latent reasoning (Chen et al., 2026c; Zhu et al., 2025c)	Latent representations, horizontal and vertical recurrence, training, and internal mechanisms
Logical reasoning (Cheng et al., 2025a)	Logical question answering and consistency, including solver-based, prompting, and fine-tuning methods
Automated planning (Aghzal et al., 2025)	LLMs as planners or components of planning systems, with emphasis on long horizons
Inductive reasoning (Chen et al., 2025b)	Inferring rules from observations through post-training, test-time exploration, and data augmentation
Efficient reasoning (Wang et al., 2026b)	Reducing reasoning cost through post-training, CoT compression, and adaptive test-time computation
Mathematical reasoning (Wang et al., 2025)	Mathematical comprehension and answer generation through pre-training, prompting, SFT, RL, and extended CoT
Multimodal reasoning (Li et al., 2025)	Development from perception-driven pipelines to multimodal CoT, RL, and native reasoning models
Neuro-symbolic reasoning (Yang et al., 2025)	Symbolic supervision, LLM-assisted symbolic reasoning, and hybrid neural-symbolic systems
Reasoning generalization (ours)	Transfer of reasoning procedures across distribution shifts, covering training, inference, architecture, and behavioral, mechanistic, and theoretical evidence.

Table 5: Main focus of related reasoning surveys and this work.

C Further Analytical Evidence

This appendix expands each paragraph of §6 under the same heading and adds the expressivity results on which the theoretical paragraphs build. Table 6 summarizes the lines of analysis, while Table 7 lists representative behavioral diagnostics and the shifts they target.

C.1 Behavioral Evidence

Compositional generalization. The decline reported by Dziri et al. (2023) was measured on multiplication, logic-grid puzzles, and dynamic programming. On formal languages, performance depends on whether a prompt stays within the rules learned in training (Mészáros et al., 2024). Accuracy on MathGAP falls as proof trees become deeper, wider, and nonlinear (Opedal et al., 2025), and XDomainBench reports a collapse as more scientific domains are combined, partly because each added domain also makes the problem harder (Gong et al., 2026). Since both benchmarks enlarge the combination together with its depth or difficulty, neither isolates composition; two-hop questions come closer to doing so because the number of steps is fixed. Scale alone does not provide systematicity. On Compositional-ARC, a 5.7M-parameter model meta-trained for compositionality generalizes to unseen combinations of geometric transformations where frontier LLMs do not (Mondorf et al., 2025).

Length, depth, and difficulty. These axes can diverge within a single environment. In a controlled shortest-path setting, models transfer to unseen maps but fail as the planning horizon grows, which the authors attribute to recursive instability (Tong et al., 2026). Length transfer is usually partial. Scratchpads improve extrapolation on algorithmic tasks without removing the underlying failure (Anil et al., 2022), and the seed dependence reported by Zhou et al. (2024d) is consistent with underspecification: the same data admit both an extrapolating solution and a shortcut, and each training run settles on one of them (§6.3). Length generalization can also transfer between tasks, as training on a longer, related auxiliary task helps a target task extrapolate beyond its own training lengths (Cai et al., 2025). For difficulty, ordering deductive-reasoning data from easy to hard gives no robust advantage over random order under either SFT or RL (Mordig et al., 2026).

Prior exposure. In chess, both performance and the benefit of each additional reasoning token decline as relevant priors become sparse (Pleiss et al., 2026). When a hard perturbation invalidates the original solution, models keep applying the memorized technique, and they do so more often when the original problem appears in context (Huang et al., 2025a). Tasks designed to fall outside pretraining, such as programming in esoteric languages, reveal failures that familiar benchmarks hide (Sharma and Chopra, 2026). Knowledge that is present is not always used. Models that cannot reverse a fine-tuned fact succeed when the same fact is given in context (Berglund et al., 2024), and models can acquire abstract structure yet make only limited use of it for compositional generation at test time (Chen et al., 2026b).

Key takeaways

- **Transfer is axis-specific.** Success on one named shift does not predict success on another.
- **Knowing is not composing.** Stored components can fail to combine, even at scale.
- **"Unseen" is stage-relative.** Apparent generality depends on the reference-exposure stage.

C.2 Mechanistic Evidence

Shared circuits. de Varda et al. (2026) localize their circuits with attribution patching. In small transformers, causal interventions can recover compact RASP subprograms that suffice to explain length-generalizing models (Huang et al., 2026b), so reusable computation can be located directly instead of being inferred from scores. Reuse has identifiable limits. Whether pattern matching succeeds on an instance is predicted by how many training contexts witness the relevant functional equivalence, and when a variable reaches the output through several paths, models fail to form a single intermediate representation (Chang et al., 2025). Together with the failure on newly integrated knowledge noted in §6.2 (He et al., 2026), these limits mean that overlap measured on familiar data supports a transfer claim only if it also holds on the shifted split.

Interfaces between steps. In transformers trained from scratch on two-hop queries, lower layers build a representation of the bridge entity, but when the second hop departs from its training

distribution, upper layers map that representation to outputs instead of reasoning over it (Zhang et al., 2026b). For facts injected by fine-tuning, relocating their internal representations by self-patching substantially improves failed cases, which is consistent with the facts being stored but not routed to the layers where later computation uses them (Dai et al., 2026). In small transformers trained from scratch on SCAN, a homomorphism error measured on hidden states predicts compositional transfer under noise injection, and regularizing this error during training improves the transfer (An and Du, 2026). Because the regularizer is a training intervention, this result goes beyond correlation, although it may change more than the measured property. Read alongside the two-hop results in §6.2, these findings locate the bottleneck at the hand-off between operations rather than in the presence of the facts or subskills.

Learning dynamics. With extended training, vanilla transformers come to generalize hierarchically on structurally novel inputs long after in-domain accuracy has saturated (Murty et al., 2023). A grokked rule can nevertheless remain bound to the facts seen in training (Wang et al., 2024a), which mirrors the limit of circuit reuse on new knowledge. Regularization can act within a critical period: on a controlled compositional task, weight decay applied during a single window of training matches the out-of-distribution accuracy of weight decay applied throughout training (Ali, 2026). In LLM post-training, excessive SFT can erode the plasticity that later RL needs (§3.3.3; Liu et al., 2026c). Each of these findings is tied to particular models, seeds, and finite test ranges, and theory is needed to separate such regularities from guarantees.

Key takeaways

- **Overlap predicts transfer.** Shared circuits predict transfer across variants, yet can fail on new knowledge.
- **Failures sit at interfaces.** Computed intermediate results may not reach the next step.
- **Optimization selects the solution.** Training length and the evaluated checkpoint shape what transfers.

C.3 Theoretical Evidence

Expressivity. Learnability results presuppose that the model class can represent the target procedure, and what it can represent depends on the computation budget and the state encoding. The number of intermediate steps is itself a resource. Under assumptions on layer normalization, transformer decoders with linearly many decoding steps recognize all regular languages, and with polynomially many steps they recognize exactly the problems solvable in polynomial time (Merrill and Sabharwal, 2024). Without intermediate steps, constant-depth transformers with constant-bit precision are limited to AC^0; with as many chain-of-thought steps as the size of a Boolean circuit, they can evaluate that circuit (Li et al., 2024c). Extrapolating constructions further depend on how state is encoded. A universal-transformer construction provably evaluates Boolean circuits of any length and depth, but it assumes fully parenthesized inputs, a positional encoding that tracks gate depth, and a number of iterations that grows with depth (Ito et al., 2026). Such constructions do not carry over directly to finite-precision LLMs, and representability alone does not show that training will find them.

Learnability. Scratchpads whose content is unsupervised do not break the globality barrier of Abbe et al. (2024), and the vocabulary of the trace matters as well. Under standard positional encodings and a finite alphabet, chain-of-thought learning that generalizes in length cannot go beyond TC^0, whereas signpost tokens, whose vocabulary grows with problem size, allow a length-generalizable simulation of Turing machines (Kraus et al., 2026). Decomposition can reduce sample complexity. When the learner receives feedback on intermediate states, a recursive curriculum for simulating semiautomata over T steps reduces the required supervision from $\Omega(T)$ to $2^{O(\sqrt{\log T})}$ tokens (Rajaraman et al., 2026). Empirically, ablations in tiny transformers trained on Dyck languages attribute the gain from intermediate traces to their content rather than to the extra tokens (Valle and Reid, 2026). These results suggest why scratchpads give only partial gains under length shift: a trace helps when it turns a global dependency into local, reusable updates, and otherwise it adds tokens without changing what must be learned.

Other results concern inductive bias, which determines which of several fitting solutions training reaches. Beyond the logarithmic-depth construction in §6.3, Liu et al. (2023) show that solvable semiautomata can be simulated with depth independent of T. Zhou et al. (2024a) conjecture,

with experimental support, that transformers length-generalize on tasks that have a short program in RASP-L, a restricted language of transformer-compatible operations. On synthetic state-tracking tasks, Huang et al. (2025c) prove that gradient descent learns chains of thought whose length extrapolation is governed by the algebraic structure of the task, through attention concentration. Kazemnejad et al. (2023) prove that a decoder without positional encoding can represent both absolute and relative positions, and they observe that after training its attention mostly resembles a relative scheme and extrapolates better than explicit encodings. In the retrieval task of Singh et al. (2026), rotary encodings make attention logits depend only on the relative offset, whereas a head with learned absolute encodings stays fixed to a position inside the training range. Data matter as well, but they are not sufficient on their own: the guarantees of Ahuja and Mansouri (2024) for length and compositional transfer require both diverse training support and constraints that rule out direct memorization.

Certification. The negative result in §6.3 concerns C-RASP, a formalism linked to length generalization, so its reach depends on what C-RASP can express; for regular languages, this is determined by their algebraic decomposition (Yang et al., 2026b). For the positive fragment of C-RASP, which is equivalent to fixed-precision transformers, a computable length generalization bound exists but is necessarily exponential (Yang et al., 2026a). For causal transformers with learnable absolute positional encodings, Huang et al. (2025b) characterize the functions that an idealized inference scheme with a norm-based regularizer identifies in the limit. This characterization proves that length generalization is possible for a rich family of problems, and in experiments it predicts success and failure across algorithmic and formal-language tasks. The positive result of Izzo et al. (2026) in §6.3 also bounds the training length it requires. Plan verification provides a positive case for checking rather than generating: transformers provably learn to verify long plans in a large class of classical planning domains, even as the vocabulary grows with the number of objects (Sarrof et al., 2026).

Key takeaways

- **Representable is not learnable.** Expressible procedures need not be learned.
- **Locality and bias govern selection.** Global targets resist learning without local steps, and standard training can prefer brittle shortcuts.
- **Finite tests do not certify.** Accuracy on a finite range is evidence about that range alone.

D Datasets and Evaluation Metrics

This appendix describes the resources behind the evidence in the main text: the supervision a training dataset provides, the shift an evaluation set implements, and the quantity a reported score measures. Tables 9 and 13 list the datasets, and Appendix D.3 documents the comparison of hybrid methods in §3.3.3.

D.1 Datasets

D.1.1 Datasets for Training

Supervision. Training datasets differ in the supervision they provide. Prompts with verifiable answers support outcome-reward training without demonstrated solutions: DAPO-Math rewrites problems to obtain integer answers, and DeepScaleR, ORZ-Math, and Big-Math enlarge the pool of verifiable prompts (Yu et al., 2025; Luo et al., 2025; Hu et al., 2025a; Albalak et al., 2025). GSM8K and MATH include human-written solutions and can support either supervised or reward-based training (Cobbe et al., 2021; Hendrycks et al., 2021). Demonstration corpora supply reasoning traces to imitate. NuminaMath-CoT contains reformatted solutions, OpenR1 pairs NuminaMath 1.5 problems with several DeepSeek-R1 traces (Li et al., 2024a; Hugging Face, 2025), and LIMO and s1K select small sets of 817 and 1,000 demonstrations, respectively (Ye et al., 2025b; Muennighoff et al., 2025). These categories describe the supervision a dataset offers rather than its only possible use: a trace corpus can supply prompts for RL, and a generator can produce supervised traces.

Controllable distributions. Knights and Knaves, Reasoning Gym, SynLogic, and Enigmata provide generators and verifiers with adjustable task parameters (Xie et al., 2025; Stojanovski et al., 2025; Liu et al., 2025a; Chen et al., 2025a), so training and test sets can be separated deliberately by length, structure, or difficulty. A held-out random seed alone yields new instances, not a structural shift.

Line of analysis	Question answered	Main finding	Interpretation limit
Behavioral evidence (§6.1)			
Compositional generalization	Do familiar primitives transfer to new combinations?	Loss grows with the depth, width, and order of the combination	Template overlap can make recombination look like composition
Length, depth, and difficulty	Does the procedure continue beyond the trained range?	Transfer is partial, axis-specific, and sensitive to the training seed	A finite extrapolation range is not unbounded extrapolation
Prior exposure	How far does the test lie from earlier training exposure?	Benefits shrink as priors become sparse, conventions change, or facts are queried in the unseen direction	Pre-training exposure is inferred, not observed
Mechanistic evidence (§6.2)			
Shared circuits	Do successes on both sides of a shift share a computation?	Circuit overlap predicts format transfer and item-level correctness	Overlap on familiar data need not hold on the shifted split
Interfaces between steps	Can each step read the result of the previous one?	Bridge entities are bypassed, resolved too late, or stored out of reach; homomorphism error predicts transfer	Repairing patches are selected per case; training interventions may change more than the measured property
Learning dynamics	Which of several fitting solutions does training select?	Rules emerge late, can stay bound to seen facts, and depend on regularization timing	Tied to specific models, seeds, and checkpoints
Theoretical evidence (§6.3)			
Learnability	Will training on finite data select a procedure the model class can represent?	Training finds brittle shortcuts unless locality, diverse support, or symmetry favor extrapolating solutions	Representability does not imply learnability; assumptions rarely match LLM training
Certification	Can a finite test establish extrapolation?	Finite training lengths suffice for restricted classes; no computable test length covers all two-layer C-RASP programs	Positive results assume idealized training or cover verification only

Table 6: Lines of analysis: the question each answers, its main finding, and the limit on interpreting it. Rows follow the paragraph headings of §6. Behavioral rows rest on performance under a named shift, mechanistic rows on representational or interventional evidence, and theoretical rows on results under explicit assumptions.

Decontamination serves a different purpose: removing benchmark overlap, as in DeepMath, reduces leakage but does not define an OOD split (He et al., 2025b). Multi-domain resources such as GURU and WebInstruct-verified support source-to-target studies, and DataFlex-RL fixes a mixture of math, logic, and science data to compare data policies (Cheng et al., 2025c; Ma et al., 2025b; Liang et al., 2026). For these resources, domain coverage and verifier reliability should be reported separately.

Counting units. Table 9 counts problems unless stated otherwise, because problems, traces, and task types are not comparable units. OpenR1-Math-220k, for example, contains 225,129 problems in its full release (93,733 in the default configuration and 131,396 in the extended one), each with two to four traces (Hugging Face, 2025). The hybrid comparisons in §3.3.3 use OpenR1-Math-46k-8192, a subset of the default configuration with correct, short traces (Yan et al., 2025; Limozin et al., 2026).

D.1.2 Datasets for Evaluation

Table 8 summarizes the main distribution shifts used in reasoning evaluations.

Held-out benchmarks. Mathematical post-training is commonly evaluated on AIME, AMC, MATH-500, Minerva/OCW, and OlympiadBench, with ARC-Challenge, GPQA-Diamond, and MMLU-Pro as checks outside mathematics (Yan et al., 2025; Limozin et al., 2026). We refer to these as the mathematics and out-of-domain suites, although even the mathematical tests may differ from the training data in difficulty, subject, or source. SuperGPQA, HLE, ZebraLogic, BBH, HumanEval, and LiveCodeBench broaden the range of content (Du et al., 2025; Phan et al., 2026; Lin et al., 2025; Suzgun et al., 2023; Chen et al., 2021; Jain et al., 2024). Whether a benchmark is ID or OOD depends on the model's pre-training, its adaptation data, and the task definition, so the label cannot be assigned to the benchmark alone.

Controlled shifts. Controlled benchmarks aim to change one property while holding the others

Diagnostic	Primary shift	Main observation	Interpretation limit
GSM-Plus (Li et al., 2024b)	Statements and answer targets	Correctness is brittle even on variants of solved problems	Perturbations remain within grade-school arithmetic
GSM-Symbolic (Mirzadeh et al., 2025)	Numeric instantiation and distractor clauses	Equivalent templates yield unstable accuracy	Template generation cannot test every skill
MathGAP (Opedal et al., 2025)	Proof-tree depth, width, and shape	Accuracy declines with structural complexity	Generated word problems simplify language variation
Compositional-ARC (Mondorf et al., 2025)	Unseen transformation combinations	Meta-learned small models can exceed frontier LLM systematicity	Abstract spatial tasks are controlled but narrow
General365 (Liu et al., 2026a)	Domain and variant	Broad reasoning remains below specialized math performance	Variant construction may not remove familiarity
XDomainBench (Gong et al., 2026)	Cross-domain composition	Failure grows with composition order and interaction	Knowledge and reasoning are partly entangled
Numeric remapping (Barker et al., 2026)	Surface-preserving numeric change	Robustness differs sharply by dataset structure	Arithmetic word problems only
Elenchos (Steiglechner et al., 2026)	Abductive rule mutation	Detection is stronger than causal attribution	Formal systems are controlled but narrow
OOD visual planning (Neuhaus et al., 2026)	Map size and representation	CoT helps ID more than most OOD settings	Simple navigation does not span open-world planning
Chess generalization (Pleiss et al., 2026)	Density of useful priors	Performance falls as relevant priors become sparse	Training exposure is inferred, not observed
OMEGA (Sun et al., 2025)	Mathematical novelty type	Separates exploratory, compositional, and transformative transfer	Novelty taxonomy depends on task construction
CoT as a mirage (Zhao et al., 2025)	Data distribution	Apparent reasoning depends on distribution design	Behavioral evidence does not identify a mechanism

Table 7: Representative behavioral diagnostics: the shift each targets, its main observation, and the limit on interpreting it.

fixed. GSM-IC adds irrelevant context, GSM-Plus applies perturbations that may also change difficulty, and GSM-Symbolic samples variants from mathematical templates (Shi et al., 2023; Li et al., 2024b; Mirzadeh et al., 2025). SCAN, COGS, and CFQ hold out lengths or combinations of familiar primitives (Lake and Baroni, 2018; Kim and Linzen, 2020; Keysers et al., 2020). MathGAP varies proof depth and width, Compositional-ARC holds out geometric compositions, and OMEGA separates greater complexity from new combinations and new strategies (Opedal et al., 2025; Mondorf et al., 2025; Sun et al., 2025). CLRS-30 was developed to probe neural algorithmic executors and is not a standard LLM leaderboard (Veličković et al., 2022). EsoLang-Bench changes the programming language while keeping the algorithmic problems; the scarcity of public data in these languages is a proxy for low exposure, not proof that pretraining never included them (Sharma and Chopra, 2026). General365 derives 1,095 variants from 365 seed problems, so statistical analyses should account for the shared seeds (Liu et al., 2026a).

Recency and contamination. Examination items and rolling benchmarks support temporal checks only when the items postdate the evaluated model's relevant training cutoff (Jain et al., 2024; White et al., 2024); AIME 2025, for instance, is not automatically unseen for models trained after its release. GSM1K was withheld at its initial publication, which supported its original comparison with GSM8K, but evaluations after its public release require a new exposure audit (Zhang et al., 2024b). Neither a publication date nor exact-match decontamination excludes exposure to equivalent problems. Table 13 therefore reports the evaluation axis and dataset version of each benchmark instead of a fixed OOD label.

D.2 Evaluation Metrics for Reasoning Generalization

Reliability under a fixed decoding protocol. For N problems and a task-specific correctness checker V_q, single-sample accuracy is

$$\mathrm{Acc} = \frac{1}{N}\sum_{q=1}^{N} V_q(\hat{y}_q).$$

The checker may normalize mathematical expressions, execute code, or compare labels. With k independently sampled responses per problem,

$$\mathrm{avg@}k = \frac{1}{N}\sum_{q}\frac{1}{k}\sum_{s=1}^{k} V_q(\hat{y}_{q,s}).$$

Shift axis	Train → test	What it probes
Instance	new samples, same generator	statistical generalization
Surface / format	paraphrase, renamed symbols, new notation	invariance to irrelevant form
Composition	new combinations of seen primitives	rule reuse and recombination
Length / depth	longer inputs, more hops, deeper nesting	extrapolation of iteration
Difficulty	easier or harder than training range	transfer across capability regimes
Task / domain	new task family or subject	portability of procedures
Language / modality	other language, format, image, audio	reasoning vs. interface
Environment / tool	new APIs, dynamics, rewards	agentic and procedural transfer
Knowledge	new facts, familiar rules	rule use on new content

Table 8: Axes along which reasoning evaluations shift between training and test. When several axes change together, the source of an improvement or failure becomes harder to identify.

This quantity estimates the expected single-sample accuracy under the chosen decoding settings; it does not measure success within k attempts. Maj@k instead grades the most frequent normalized answer under a stated tie-breaking rule (Wang et al., 2023), so it evaluates an aggregation procedure that spends k samples at deployment.

Coverage at a stated budget. Pass@k measures whether at least one of k samples succeeds. Given $n \geq k$ independent samples at fixed decoding settings, of which c_q are correct, the estimator is (Chen et al., 2021)

$$\text{pass@}k = \frac{1}{N} \sum_{q} \left[1 - \frac{\binom{n-c_q}{k}}{\binom{n}{k}} \right],$$

with $\binom{n-c_q}{k} = 0$ when $n - c_q < k$. Because pass@k relies on an oracle success criterion, it does not measure the accuracy of a deployable selector unless a suitable verifier is available. A higher pass@1 with unchanged or lower large-k coverage is consistent with post-training that reweights existing solutions (Yue et al., 2025), although finite sampling cannot establish with certainty whether a solution is present or absent. We therefore recommend reporting coverage curves together with sample and token budgets.

Robustness to a matched transformation. For comparable reference and shifted sets, both scores should be reported together with the absolute and relative drops,

$$\Delta = \text{Acc}_{\text{ref}} - \text{Acc}_{\text{shift}},$$
$$\Delta_{\text{rel}} = \Delta / \text{Acc}_{\text{ref}},$$

where the relative drop requires positive reference accuracy. Under the same 0–1 loss and evaluation procedure, Δ equals the transfer gap defined in §2. A smaller gap is not necessarily better, since it may result from lower reference accuracy. When variants share an original problem or template, results are best aggregated at that level. GSM-Symbolic reports variation across generated test sets (Mirzadeh et al., 2025), and GSM-IC also reports the fraction of base problems solved under *all* distractor variants (Shi et al., 2023). Conditional variant accuracy, $P(\text{variant correct} \mid \text{original correct})$, measures how often correctness is retained on the originally solved subset, but it is not a complete measure of consistency.

Length and composition. We recommend reporting length results as curves that extend beyond the training frontier and that distinguish input length, required depth, and the budget of generated tokens. As a summary, we define the largest tested length up to which accuracy stays at or above a threshold τ:

$$L^*_\tau = \max \left\{ L \in \mathcal{L} : \min_{\ell \in \mathcal{L},\, \ell \leq L} \text{Acc}(\ell) \geq \tau \right\}.$$

Here $\mathcal{L}$ is the declared grid of test lengths, and an empty set means that no tested length reaches the threshold. The summary says nothing about untested lengths. The ratio $L^*_\tau / L_{\text{train}}$ describes extrapolation relative to the final training frontier, similar to the length ratios reported in integer-addition studies (Zhou et al., 2024d). For composition, held-out structures should be compared at matched depth or length where possible (Park et al., 2025).

Transfer and retention. Transfer should be measured against the base model on each target task. For method m trained on source s, we define

$$T^{(m)}_{s \to t} = \text{Acc}_t(\theta^{(m)}_s) - \text{Acc}_t(\theta_0).$$

Dataset	Year	Domain	Problems / Tasks	Supervision
Math: Verifiable Prompts for RL				
GSM8K (train) (Cobbe et al., 2021)	2021	Grade-school math	7,473	Answer + rationale
MATH (train) (Hendrycks et al., 2021)	2021	Competition math	7,500	Answer + solution
SimpleRL-Zoo (Zeng et al., 2025)	2025	Math	∼8K per split	Answer
DAPO-Math-17k (Yu et al., 2025)	2025	Competition math	∼17K	Integer answer
DeepScaleR (Luo et al., 2025)	2025	Competition math	∼40K	Answer
DeepMath-103K (He et al., 2025b)	2025	Math	103K	Answer + traces
Skywork-OR1 (He et al., 2025a)	2025	Math / code	∼105K / 13.7K	Answer / tests
ORZ-Math (Hu et al., 2025a)	2025	Competition math	∼129K	Answer
Big-Math (Albalak et al., 2025)	2025	Math	>250K	Answer
Math: Demonstrations for SFT, Distillation, and Hybrid Training				
NuminaMath-CoT (Li et al., 2024a)	2024	Math	∼860K	CoT solution
OpenR1-Math-220k (all)† (Hugging Face, 2025)	2025	Math	∼225K	2–4 R1 traces per problem
OpenR1-Math-46k-8192†§ (Yan et al., 2025)	2025	Math	∼46K	Long CoT (R1) + answer
LIMO (released set)† (Ye et al., 2025b; GAIR, 2025)	2025	Math	817	Long CoT
s1K† (Muennighoff et al., 2025)	2025	Math, science	1,000	Long CoT
OpenThoughts releases† (Guha et al., 2025)	2025	Math, code, science	114K / 1M / 1.2M	Long CoT
Logic, Puzzles, and Procedural Environments				
Knights & Knaves‡ (Xie et al., 2025)	2025	Logic puzzles	Procedural	Assignment checker
Reasoning Gym‡ (Stojanovski et al., 2025)	2025	Algorithmic, logic	100+ generators	Task verifier
SynLogic‡ (Liu et al., 2025a)	2025	Logic	35 task types	Task verifier
Enigmata‡ (Chen et al., 2025a)	2025	Puzzles	36 task types	Task verifier
Multi-domain				
SciQ (train) (Welbl et al., 2017)	2017	Science MCQ	11,679	Correct option
GURU (Cheng et al., 2025c)	2025	Six domains	92K	Domain verifiers
WebInstruct-verified (Ma et al., 2025b)	2025	Multi-domain	∼230K	Model-based verifier
DataFlex-RL mixture (Liang et al., 2026)	2026	Math, logic, science	15K	Rule-based checks

Table 9: Summary of datasets used to train reasoning models. † Counts refer to problems, not traces; a problem may have several traces. ‡ Procedural generator; task counts are not instance counts. § Correct, short-trace subset of the OpenR1 default configuration. "∼" marks rounded counts.

A positive value on a target excluded from adaptation supports transfer, and a negative value records degradation. Sequential retention is a different quantity: it compares an earlier and a later checkpoint on domains that were already learned. Raw accuracy differences across domains also reflect task difficulty and should not be read as a normalized measure of transfer.

Aggregation, uncertainty, and cost. A domain-balanced summary averages within domains before averaging across them:

$$S_{\mathrm{DB}} = \frac{1}{K}\sum_{d=1}^{K}\frac{1}{|\mathcal{B}_d|}\sum_{b\in\mathcal{B}_d}\mathrm{Acc}_b.$$

This average assumes comparable units, so rubric credit, F1, and accuracy should not be averaged without a justified normalization. Benchmark coverage alone can reverse method rankings (Liang et al., 2026), which is why per-task scores should accompany any aggregate. Paired method differences need uncertainty estimates over matched training seeds; item-level bootstrap tests, such as those reported for LUFFY, do not capture training-seed variance (Yan et al., 2025). Bootstrap resampling of generated variants should operate on their shared problem or template. Run-level standard deviations and confidence intervals answer different questions and need to be labeled accordingly, and inference methods are compared at a fixed budget or through accuracy-cost curves (Snell et al., 2024).

D.3 Protocols and Published Results

Common reproduction study. Table 10 is taken from Limozin et al. (2026), in which the corrected Qwen2.5-Math-7B (Yang et al., 2024) baselines and the LUFFY and ReLIFT reproductions all use OpenR1-Math-46k-8192 (Hugging Face, 2025; Yan et al., 2025). AIME24, AIME25, and AMC are scored with avg@32 and the remaining tests with pass@1, using a temperature of 0.6, a maximum response length of 8,192, and shuffled options for multiple-choice questions. The corrected baselines come from three independent end-to-end

runs, whereas each reproduction is a single run. Sequential training consists of SFT followed by 500 GRPO steps (Shao et al., 2024), and the 50-step variant also raises the RL learning rate from 10^{-6} to 5×10^{-6}. The comparison therefore shares an evaluation protocol but does not match compute.

Published results. Table 11 keeps the originally reported hybrid results separate from the reproduction, and Table 12 summarizes how each method combines SFT and RL. All five methods build on GRPO-family objectives: LUFFY adds off-policy R1 traces to each GRPO rollout group, ReLIFT interleaves GRPO with SFT on questions that all rollouts fail, SRFT adds an entropy-weighted SFT loss to group-relative RL losses, Prefix-RFT applies Dr. GRPO (Liu et al., 2025c) to rollouts that continue demonstration prefixes, and HPT switches each question between SFT and a Dr. GRPO-style RL loss according to rollout accuracy. LUFFY and Prefix-RFT report the full nine-benchmark suite (Yan et al., 2025; Huang et al., 2025d). ReLIFT uses avg@8 on MATH-500 and OlympiadBench and does not report Minerva (Ma et al., 2025a), SRFT omits AIME25 (Fu et al., 2025), and HPT reports only two OOD tests, omitting MMLU-Pro, with top-p sampling (Lv et al., 2025). We leave these cells empty rather than importing values from later compiled tables. The original comparisons also differ in SFT implementation and hyperparameters, as audited by Limozin et al. (2026), so the gap between published and reproduced scores mixes implementation differences with seed variance.

Method	Math avg.	ARC-C	GPQA-D	MMLU-Pro	OOD avg.
Corrected SFT	52.2 ± 0.2	82.5 ± 0.6	24.6 ± 2.9	50.2 ± 0.3	52.4 ± 1.1
SFT→GRPO (50 steps)	55.6 ± 0.5	84.0 ± 0.5	39.6 ± 1.0	53.9 ± 0.4	59.2 ± 0.2
SFT→GRPO (500 steps)	57.0 ± 0.3	84.0 ± 0.7	40.6 ± 1.8	55.1 ± 0.6	59.9 ± 0.9
LUFFY	46.3	81.4	41.8	50.9	58.0
ReLIFT	48.8	80.7	36.7	51.3	56.2

Table 10: Accuracy (%) from Limozin et al. (2026); the originally published results are given in Table 11. All rows use Qwen2.5-Math-7B (Yang et al., 2024) and OpenR1-Math-46k-8192 (Hugging Face, 2025; Yan et al., 2025). Corrected baselines report mean ± standard deviation over three independent runs; mixed-policy reproductions use one run. Math avg. covers six benchmarks and OOD avg. covers the three displayed benchmarks. Training compute is not matched across all rows.

Method	AIME24	AIME25	AMC	MATH-500	Minerva	Olympiad	Math avg.	ARC-C	GPQA-D	MMLU-Pro	OOD avg.
LUFFY (Yan et al., 2025)	29.4	23.1	65.6	87.6	37.5	57.2	50.1	80.5	39.9	53.0	57.8
ReLIFT (Ma et al., 2025a)	28.3	22.9	65.1	87.9	—	57.3	—	81.6	43.1	53.9	59.5
SRFT (Fu et al., 2025)	35.3	—	74.3	89.8	39.7	58.3	—	85.3	46.4	55.9	62.5
Prefix-RFT (Huang et al., 2025d)	31.8	26.4	68.2	88.4	40.3	55.7	51.8	84.0	39.1	52.1	58.4
HPT (Lv et al., 2025)	33.0	21.9	69.4	89.2	46.0	56.9	52.7	81.6	42.9	—	—

Table 11: Originally reported hybrid scores (%). A shared model and nominal dataset do not imply identical evaluation or compute. ReLIFT uses avg@8 on two mathematical tests; SRFT and HPT omit different tests. Dashes denote absent original results. Means cover the complete six-math or three-OOD suite; ReLIFT's three-test OOD mean is calculated here from its Tables 1 and 5.

Method	RL algorithm	How SFT and RL are combined	Evaluation or comparison limit
Corrected SFT	None	SFT on demonstrations only; corrected implementation	Three independent runs
SFT→GRPO (50 steps)	GRPO	Full SFT, then 50 GRPO steps	Higher RL learning rate; three runs
SFT→GRPO (500 steps)	GRPO	Full SFT, then 500 GRPO steps	Three runs; extra SFT compute
LUFFY (Yan et al., 2025)	Mixed-policy GRPO	Off-policy R1 traces added to each on-policy rollout group; policy shaping of the off-policy ratio	Published or one-run reproduced results, kept separate
ReLIFT (Ma et al., 2025a)	GRPO[†]	GRPO interleaved with SFT steps on questions whose rollouts all fail	Original avg@8 on two tests; no Minerva
SRFT (Fu et al., 2025)	GRPO-style, LUFFY-like off-policy term	One stage: entropy-weighted SFT loss plus group-relative RL losses on demonstrations and own rollouts	Original table omits AIME25
Prefix-RFT (Huang et al., 2025d)	Dr. GRPO	One of eight rollouts continues a demonstration prefix; only high-entropy prefix tokens are updated	Original reported results
HPT (Lv et al., 2025)	Dr. GRPO-style GRPO[‡]	Per-question gate: SFT on the demonstration if all eight rollouts fail, RL otherwise	Two OOD tests; top-p sampling

Table 12: Settings and limits of the hybrid comparison, including the RL algorithm of each method (GRPO: Shao et al., 2024; Dr. GRPO: Liu et al., 2025c). All listed methods use Qwen2.5-Math-7B (Yang et al., 2024) and the filtered OpenR1 data (Hugging Face, 2025; Yan et al., 2025), but training allocations and evaluation details vary. [†]Without length and standard-deviation normalization, as in Dr. GRPO. [‡]The setup names GRPO and the loss definition specifies Dr. GRPO: responses are not length-normalized, but advantages keep standard-deviation normalization.

Dataset / Benchmark	Characteristics and Evaluation Use	Dataset Size / Subset	Evaluation Metric
	Mathematics		
GSM8K (Cobbe et al., 2021)	Grade-school word problems requiring multi-step arithmetic. Some studies group problems by solution-step count to probe step-count extrapolation (Zhou et al., 2023).	7,473 train / 1,319 test	Accuracy
MAWPS	Arithmetic word-problem collection; used as a zero-shot target in tool-use studies (Gao et al., 2025b).	Protocol-dependent	Accuracy
Minerva / OCW (Lewkowycz et al., 2022)	Undergraduate STEM problems from MIT OpenCourseWare with numeric or symbolic answers.	272 problems	Accuracy
MATH (Hendrycks et al., 2021)	Competition problems with worked solutions, five difficulty levels and seven subjects. Difficulty-stratified analyses (Snell et al., 2024) measure performance across difficulty, which is not by itself distribution shift.	12,500 problems (7,500 train / 5,000 test)	Accuracy (Exact Match on final answer)
MATH-500 (Lightman et al., 2024)	500-problem subset of the MATH test split, introduced with PRM800K and widely used as a fast evaluation set.	500 problems (subset of MATH)	Accuracy; pass@k in sampling studies
AMC	American Mathematics Competitions problems. Harnesses select different years, so scores are comparable only under the same selection.	40 or 83 problems, depending on harness	avg@k
AIME 2024 / 2025 / 2026	American Invitational Mathematics Examination editions with integer answers. Each edition is a separate problem set; recency acts as a contamination control only relative to a model's data cutoff.	30 per edition (AIME I and II combined)	avg@k; Accuracy / pass@1 in some studies
CMIMC 2025; BRUMO 2025	Two university-hosted competitions used as recent competition mathematics in Ma et al. (2026). Treated as separate competitions.	Protocol-dependent	avg@k (k=4 in the cited study)
OlympiadBench (He et al., 2024a)	Olympiad-level mathematics and physics problems, bilingual, with and without images.	8,476 problems in the release; 675 English text-only mathematics problems commonly used (SRFT reports 674)	Accuracy
MATH^2	Harder questions composed from skills of MATH, used to test selection under increased difficulty (Sareen et al., 2025). Difficulty scaling alone does not establish distribution shift.	Protocol-dependent	Accuracy
ProcessBench	Step-level error identification with subsets from GSM8K, MATH, OlympiadBench and Omni-MATH, used to assess process reward models rather than answer accuracy. Zhu et al. (2025a) treat the competition subsets as question-OOD relative to PRM training data. Omni-MATH also exists as a standalone benchmark and is listed here only through ProcessBench.	Subsets per study	F1 (first-error identification); ArithACC in Zhu et al. (2025a)
	Science, Knowledge, and Factuality		
ARC (Challenge / Easy) (Clark et al., 2018)	Grade-school multiple-choice science questions; ARC-Challenge is the harder split and ARC-Easy a separate split of the same corpus. ARC-C serves as a reference set in assessment-transfer studies, where AUROC is reported. Distinct from ARC-AGI.	ARC-Challenge: 1,119 train / 1,172 test	Accuracy
GPQA-Diamond (Rein et al., 2024)	Expert-written graduate-level science questions; the Diamond subset keeps those validated most strictly.	198 questions	Accuracy; pass@k in some studies
MMLU-Pro (Wang et al., 2024b)	Broad-subject multiple-choice questions with more answer options and more reasoning-heavy items than MMLU.	12,032 test questions	Accuracy

Continued on next page

Table 13 (continued)

Dataset / Benchmark	Characteristics and Evaluation Use	Dataset Size / Subset	Evaluation Metric
SuperGPQA (Du et al., 2025)	Graduate-level questions across 285 disciplines.	26,529 questions	Accuracy
Humanity's Last Exam (Phan et al., 2026)	Expert-level closed-ended questions across many fields; a difficulty benchmark, not a controlled shift.	2,500 questions	Accuracy
ScienceArena (Zhao et al., 2026a)	Questions from recent scientific olympiad competitions; temporal validity depends on model cutoff.	13 competitions	Rubric credit
C-Eval Hard	Hard subset of C-Eval (Chinese-language subject examinations), used as a non-source benchmark in Zhang et al. (2024a).	Protocol-dependent	Accuracy
FACTOR (Muhlgay et al., 2024)	Factuality benchmarks generated from a text corpus by perturbing facts; the model should prefer the factual completion. Wiki- and News-based variants exist.	Generated; per-variant sizes in the source	Accuracy
HaluEval-Summary	Summarization subset of HaluEval for detecting hallucinated summaries; used in evaluation of divide-and-conquer prompting (Zhang et al., 2024c).	Protocol-dependent	Accuracy; F1; Precision; Recall
SciFact (misinformation variant)	Author-constructed paragraph-level misinformation variant of SciFact; results depend on the construction in Zhang et al. (2024c).	Variant-specific	F1; Precision; Recall
	Logic, Puzzles, and Code		
ARC-AGI (Chollet, 2019)	Grid-transformation puzzles requiring adaptation to unfamiliar patterns from few examples (ARC-AGI-1). Distinct from AI2 ARC and Compositional-ARC.	400 training / 400 public evaluation tasks	Accuracy (exact output match)
BBH (Suzgun et al., 2023)	Challenging BIG-Bench tasks spanning symbolic, linguistic and logical reasoning.	23 tasks	Accuracy
T4D	Theory-of-mind task of choosing actions from inferred mental states; used in Self-Discover evaluation (Zhou et al., 2024b).	Protocol-dependent	Accuracy
HumanEval / HumanEval+ (Chen et al., 2021)	Python function-synthesis problems checked by unit tests. HumanEval+ keeps the same problems with expanded tests.	164 problems	pass@k
MBPP / MBPP+	Short Python programming tasks; MBPP+ is a variant with more extensive tests.	974 problems; a 500-problem test split is commonly used; MBPP+ uses a filtered subset	pass@k
LiveCodeBench (Jain et al., 2024)	Continuously updated coding problems; contamination control only if problems postdate the model's cutoff.	Rolling release	pass@1
Knights & Knaves (Xie et al., 2025)	Procedurally generated logic puzzles with adjustable difficulty; train and test separation depends on the generator settings used.	Procedural	Accuracy
ZebraLogic (Lin et al., 2025)	Logic-grid puzzles with controllable size, used to study scaling of multi-step constraint reasoning.	1,000 puzzles	Accuracy
	Surface Perturbation and Distractors		
GSM-IC (Shi et al., 2023)	GSM8K-style problems with inserted irrelevant context.	58,052 examples	Micro / Macro Accuracy
GSM-Plus (Li et al., 2024b)	Eight perturbation types applied to GSM8K seed problems; some perturbations also change difficulty.	10,552 variants (1,319 seeds $\times$ 8)	Accuracy
GSM-Symbolic (Mirzadeh et al., 2025)	Template-based symbolic variants of grade-school problems.	100 templates $\times$ 50 instances per variant set	Mean $\pm$ Std.
SVAMP	Word problems obtained by perturbing simple arithmetic problems; used as a zero-shot target in Gao et al. (2025b).	Protocol-dependent	Accuracy
	Composition, Length, and Systematic Generalization		

Continued on next page

Table 13 (continued)

Dataset / Benchmark	Characteristics and Evaluation Use	Dataset Size / Subset	Evaluation Metric
SCAN (length split) (Lake and Baroni, 2018)	Command-to-action mapping; the length split tests longer target sequences than seen in training.	16,990 train / 3,920 test	Exact Match
CFQ (Keysers et al., 2020)	Semantic-parsing benchmark with splits that hold out compositions of familiar parts.	239,357 examples; splits derived	Exact Match
COGS (Kim and Linzen, 2020)	Sentence-to-logical-form mapping that tests novel combinations of familiar words and structures.	24,155 train / 3,000 dev / 21,000 generalization	Exact Match
CommaQA-E	Synthetic multi-hop question answering over fictional entities; used in the Skills-in-Context evaluation (Chen et al., 2024).	Protocol-dependent	Exact Match
Last-letter concatenation (Zhou et al., 2023)	Synthetic task of concatenating the last letters of a word list; evaluated at increasing list lengths.	Per-length test sets; protocol-dependent	Accuracy
CLRS-30 (Veličković et al., 2022)	Thirty classical algorithms with graph and sequence inputs; the benchmark tests larger input sizes than training. Developed for neural algorithmic reasoning, not as a standard LLM leaderboard.	1,000 train trajectories per algorithm; test counts depend on output granularity	Micro-F1
MathGAP (Opedal et al., 2025)	Generated word problems with controllable proof depth and width for out-of-distribution evaluation.	Generated	Accuracy
Multi-digit arithmetic (Zhou et al., 2024d; Malach et al., 2026)	Addition and multiplication across digit lengths. Extrapolation is established only when training and test lengths are stated; Malach et al. (2026) evaluate up to 1,000 digits.	Protocol-dependent	Accuracy; Exact Recovery
Dynamic programming: maximum non-adjacent sum (Chen et al., 2024)	Sequence optimization evaluated on longer inputs than those in the demonstrations.	Protocol-dependent	Exact Match; Accuracy
Compositional-ARC (Mondorf et al., 2025)	ARC-derived meta-learning episodes testing systematic generalization to held-out geometric compositions.	82,908 train / 8,546 test episodes	Accuracy
OMEGA (Sun et al., 2025)	Generated math problems separating exploratory, compositional and transformative generalization.	Generated	Accuracy
XDomainBench (Gong et al., 2026)	Scientific knowledge composition across domains, used to diagnose reasoning collapse.	8,598 sessions	Recall; F1
Reversal Curse (Berglund et al., 2024)	Tests whether a relation learned in one direction ("A is B") supports the reverse ("B is A") after fine-tuning on synthetic facts.	Generated	Exact Match
Tool Use, Planning, Agents, and Modality Shifts			
A-OKVQA (Schwenk et al., 2022)	Visual question answering requiring image content and outside knowledge; a modality shift when the text-only model is the reference.	Official splits; subset sizes vary by study	Direct-answer accuracy; multiple-choice accuracy
Tower of Hanoi	Planning puzzle whose disk count controls solution length ($2^n - 1$ moves). Protocols may supply a strategy (Meyerson et al., 2025) or test extrapolation to more disks (Malach et al., 2026).	Protocol-dependent	Zero-error completion; Accuracy
GSM8K-XL	Arithmetic word problems with large numbers, used to evaluate tool selection under magnitude shift (Alazraki and Rei, 2025).	Protocol-dependent	Accuracy
FuncQA	Numeric questions requiring chains of arithmetic function calls.	Protocol-dependent	Accuracy
MINTQA	Multi-hop question answering over new and long-tail knowledge, used for tool-use decisions (Qian et al., 2025).	Protocol-dependent	Accuracy; tool-call metrics
SMART task families (Math, Time, Intention)	Task families used to evaluate tool-use restraint and cross-task transfer (Qian et al., 2025). The relation of Intention to IN3 is unconfirmed.	Protocol-dependent	Accuracy; tool-overuse rate
WebArena (Zhou et al., 2024c)	Interactive tasks in simulated websites; a cross-site subset ("Multi") is used in memory-accumulation studies (Ouyang et al., 2026).	812 tasks	Success Rate

Continued on next page

Table 13 (continued)

Dataset / Benchmark	Characteristics and Evaluation Use	Dataset Size / Subset	Evaluation Metric
Mind2Web (Deng et al., 2023)	Web-navigation tasks across websites and domains, with cross-task, cross-website and cross-domain test sets.	Protocol-dependent	Success Rate (step and task level)
SWE-bench Verified	Real GitHub issues validated as solvable, a 500-instance subset of SWE-bench (Jimenez et al., 2024). Task difficulty alone does not establish shift.	500 instances	Resolve rate
Recent and Contamination-Limited Benchmarks			
GSM1K (Zhang et al., 2024b)	Newly written grade-school problems matched to GSM8K, withheld at release to audit contamination. After public release, evaluation requires a fresh exposure audit.	1,205 problems	Accuracy
LiveBench (White et al., 2024)	Regularly refreshed questions designed to limit contamination; useful only relative to a known model cutoff.	Periodic updates	Accuracy
General365 (Liu et al., 2026a)	Seed tasks with generated variants; analyses should account for shared seeds.	365 seeds + 1,095 variants	Accuracy
EsoLang-Bench (Sharma and Chopra, 2026)	Programming problems in esoteric languages; scarce public data is a proxy for low exposure, not proof of none.	80 per language	Accuracy

Table 13: Evaluation benchmarks referenced in this survey, grouped by evaluation setting. Counts and metrics follow the cited release or the studies that report them; "Protocol-dependent" means the value is set by individual studies. Whether an evaluation is in-distribution or out-of-distribution depends on a model's training exposure under a specific protocol, not on the benchmark name.

Method / paper	Evaluation evidence	Reported measurements	Generalization evidence and caveat
Trajectory restructuring			
Least-to-Most prompting (L2M) (Zhou et al., 2023)†	Last-letter/list concatenation, SCAN length splits, and GSM8K grouped by solution-step count.	Accuracy by list length or step count.	**E1, qualified.** Evaluates structural extrapolation relative to shorter in-context exemplars. Extrapolation across semantic domains is not assessed.
Divide-and-Conquer (DaC) prompting (Zhang et al., 2024c)	HaluEval-Summary, a constructed SciFact misinformation variant, and long-integer arithmetic tasks.	Accuracy, F1, precision, recall.	**E4.** Evaluates task performance without an explicit ID/OOD comparative framework.
MAKER (Meyerson et al., 2025)	Tower of Hanoi with a supplied strategy, including a 20-disk zero-error system run.	Zero-error completion; projected per-model error and cost estimates.	**E4, provisional.** Evaluates performance on scaled tasks; a formalized ID/OOD split is not established.
SKiC (Chen et al., 2024)†	Synthetic structural tasks (last-letter concatenation, arithmetic digit-length extrapolation, dynamic programming) and standard reasoning evaluations (CommaQA-E, MATH).	Accuracy/Exact Match by task size and prompt condition; internal-skill activation rates.	**E1, qualified.** Demonstrates structural length/depth extrapolation on synthetic settings. Standard reasoning benchmark evaluations lack a paired ID/OOD split.
Constraint-First Reasoning (CFR) (Ma et al., 2026)	AIME24/25/26, CMIMC25, BRUMO25, and OlympiadBench.	Pass@1 averaged over four runs (`avg@4`); accuracy and token overhead.	**E3.** Reports quantitative gains and format robustness on competition math tasks without a predefined ID/OOD baseline.
Self-Discover (Zhou et al., 2024b)	BBH, T4D, and MATH, including transfer across model families.	Task-level accuracy gains.	**E4.** Evaluates task performance and cross-model transfer; a data-distribution OOD test is not established.
Self-Demos (He et al., 2024b)†	Query-aware demonstrations and the OOD-Tool set.	Task accuracy compared with few-shot baselines.	**E2/E3, terminology-sensitive.** Evaluates out-of-demonstration capabilities; does not explicitly test shifts from the model's pre-training distribution.
Compute scaling			
ReST-MCTS* (Zhang et al., 2024a)†	MATH-sourced training and evaluation settings compared with non-source benchmarks (GPQA-Diamond, CEval-Hard).	Accuracy and average scores for PRM-guided MCTS and self-training.	**E1*, qualified.** Reports comparative gains on non-source benchmarks. Base-model pre-training exposure is unquantified.
Boosting PRMs with MCTS (Chan et al., 2025)	Math, code, and knowledge benchmarks evaluating MCTS-generated training data and inference scaling (Best-of-N, beam search).	Benchmark scores and scaling trends under sample/search budgets.	**E3.** Evaluates scaling behavior and improvements from generated training data without a controlled ID/OOD test.
Compute-optimal test-time scaling (Snell et al., 2024)	MATH test set stratified into difficulty bins to evaluate search and sequential revision strategies.	MATH accuracy versus generation/compute budget.	**E4.** Stratifies evaluation by task difficulty; a formal ID/OOD distribution split is not constructed.
Test-Time Scaling via Error Localization (TTEL) (Chitale et al., 2026)	MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench evaluating localized regeneration.	Pass@k and token budgets under rollout/sample constraints.	**E4.** Evaluates pass@k improvements under localized revision; explicit train/test distribution shifts are not assessed.
State assessment			
Residual-stream trajectory detector (Damirchi et al., 2026)†	ARC-C (reference) and other reasoning/commonsense datasets (targets) used for cross-benchmark candidate detection.	AUROC and correct-candidate detection rates.	**E1, assessment transfer.** Evaluates transferability of the assessment/detection signal to held-out benchmarks.

Continued on next page

Table 14 (continued)

Method / paper	Evaluation evidence	Reported measurements	Generalization evidence and caveat
UPAIR (Yan et al., 2026)	Diagnostic experiments and selective interventions evaluated on MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench.	Final accuracy/token cost; partial diagnosis and intervention metrics.	**E4, provisional.** Evaluates diagnostic signals and intervention benefits without a controlled OOD comparison.
Thought Calibration (Wu et al., 2025)†	Hidden-state probes and risk-controlled halting evaluated on target sets (AIME24, GPQA, MATH-500), using s1K-1.1 as reference data.	Probe AUROC, stopping/risk statistics, task accuracy, and step/token savings.	**E1, qualified.** Assesses calibration and task gains on target sets; calibration stability under distribution shift is measured separately.
Online Reasoning Calibration (ORCA) (Zhou et al., 2026)†	Test-time online calibration evaluated on MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench.	Accuracy, generated tokens, and risk/calibration statistics.	**E1, qualified.** Explicitly evaluates calibration under target settings.
Generative verifier / RL^{V} (Sareen et al., 2025)†	Candidate selection evaluated on MATH-500, harder MATH^2, and a physics set following MATH training.	Candidate-selection accuracy under parallel test-time scaling.	**E1, task generalization only.** Demonstrates downstream selection capabilities on math-adjacent and physics tasks.
Verifier-Driven Sample Selection TTT (VDS-TTT) (Moradi et al., 2025)	High-confidence outputs used as pseudo-labels for generator adaptation on GSM8K, MATH-500, and AIME.	Exact Match/accuracy against a frozen base and verifier-only baselines.	**E3.** Evaluates target-task gains from generator adaptation; a paired reference distribution is not defined.
Retrieval-Augmented PRM (RetrievalPRM) (Zhu et al., 2025a)†	ProcessBench (GSM8K, MATH, OlympiadBench, Omni-MATH) evaluated using a PRM800K and Math-Shepherd retrieval pool.	ProcessBench F1 and ArithACC for first-error detection.	**E2, qualified.** Competition subsets are evaluated as question-OOD relative to PRM training.
Inference-time externalization			
Interactive tool-use memory for bounded-state models (Malach et al., 2026)†	Interactive memory tools evaluated on synthetic addition/multiplication (up to 1,000 digits), logical graphs, and Tower of Hanoi.	Exact-recovery and task accuracy by sequence size.	**E1, structural length generalization only.** Evaluates structural length generalization for specific architectures; semantic or algorithmic OOD is not tested.
Chain-of-Abstraction (CoA) (Gao et al., 2025b)†	Zero-shot evaluations on math datasets (GSM8K, SVAMP, MAWPS) and knowledge-intensive QA.	Exact Match/accuracy, arithmetic error rate, and latency.	**E1, provisional cross-task transfer.** Evaluates transfer to designated target datasets. Dataset overlap with pre-training is not assessed.
TECTON (Alazraki and Rei, 2025)†	Tool selection evaluated on GSM8K-XL and designated math sets with magnitude/dataset variation.	Final task accuracy.	**E1, provisional.** Demonstrates tool-selection transfer on math word problems featuring magnitude shifts.
SMART / SMARTAgent (Qian et al., 2025)†	Tool-use decisions evaluated across task types, including GSM8K, MINTQA, and Math/Time-style tasks.	Task accuracy, tool calls, tool-overuse rate, and decision confidence.	**E1, provisional.** Evaluates tool-use restraint and cross-task transfer.
ReasoningBank (Ouyang et al., 2026)	Task-stream memory accumulation evaluated on Mind2Web, WebArena Multi, and SWE-Bench-style tasks.	Success rate, average interaction steps, and token cost.	**E3.** Demonstrates strategy-level memory and within-stream improvements; an isolated held-out target is not established.
MILES (Tong and Gong, 2026)	Modular sub-instruction selection evaluated on MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench.	Majority-vote accuracy, sub-instruction rates, memory size, and accuracy–cost frontier.	**E3 for auxiliary transfer checks; E4 for main results.** Evaluates accuracy–cost trade-offs; cross-domain memory transfer is not directly assessed.

Table 14: Evaluation evidence for inference-time methods. The table summarizes the evaluated tasks and protocols, reported measurements, and strength and limitations of the generalization evidence. The family headings group methods by their primary inference-time intervention. Benchmark transfer, difficulty scaling, and performance gains alone do not establish out-of-distribution (OOD) generalization; evidence must be interpreted in light of the evaluation protocol and known training exposure.

Metric	Definition	Interpretation	Comparability considerations
Task performance			
Accuracy / Exact Match	For N examples, $\mathrm{Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\hat{y}_i = y_i]$. Exact Match applies the benchmark's answer-matching rule.	Measures the proportion of correct outputs; higher is better.	Requires compatible answer parsing, normalization, and test subsets.
Success Rate	For N tasks, $\mathrm{SR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\text{task } i \text{ succeeds}]$.	Measures the proportion of tasks completed successfully.	The benchmark's task-level success criterion must be consistent.
Sampling and selection protocols			
pass@k	Given $n_i \geq k$ sampled candidates and c_i correct candidates, $\widehat{\text{pass@}k} = \frac{1}{N}\sum_{i=1}^{N}\left(1 - \frac{\binom{n_i - c_i}{k}}{\binom{n_i}{k}}\right)$. (Chen et al., 2021)	Estimates the probability that at least one of k sampled candidates is correct; non-decreasing in k.	Match k, candidate counts, decoding settings, and compute budget.
Majority voting (maj@k)	Selects the most frequent normalized answer among k sampled candidates and evaluates the selected answer (Wang et al., 2023).	Measures the effectiveness of answer aggregation across samples.	Depends on sample count, answer normalization, and tie-breaking rules.
Best-of-N selection	Generates N candidates, selects one using a scorer or verifier, and evaluates the selected output with a task metric.	Measures task performance under a candidate-selection protocol.	A selection protocol rather than a standalone metric. Specify N and the selection procedure.
Verification and discrimination			
F1	$\mathrm{F1} = 2\frac{\mathrm{Precision}\cdot\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}}$.	Summarizes precision and recall; higher is generally better.	Depends on the positive-class definition and averaging method.
AUROC	Area under the receiver operating characteristic curve; measures how well a score ranks positive instances above negative ones.	Higher values indicate better discrimination.	Does not measure probability calibration; specify the positive class.
Efficiency and resource cost			
Tokens, latency, and compute budget	Operational costs such as generated tokens, wall-clock time, model calls, and allocated computation.	Characterizes the resources required to obtain a result.	Compare costs alongside task performance under compatible hardware, settings, and budgets (Snell et al., 2024).
Survey-defined quantities for generalization analysis			
Generalization gap Δ	$\Delta = \mathrm{Acc}_{\mathrm{ref}} - \mathrm{Acc}_{\mathrm{shift}}$, computed on matched reference and shifted evaluation sets.	A positive value indicates a performance drop under the shift.	Requires a comparable model, checkpoint, metric, and evaluation protocol. Interpret alongside both scores.
Relative drop Δ_{rel}	$\Delta_{\mathrm{rel}} = \Delta/\mathrm{Acc}_{\mathrm{ref}}$, defined when $\mathrm{Acc}_{\mathrm{ref}} > 0$.	Expresses the performance drop relative to reference accuracy.	Report alongside the absolute gap and both underlying scores.

Gain over baseline G	$G = \mathrm{Score}_{\mathrm{method,target}} - \mathrm{Score}_{\mathrm{baseline,target}}$ under the same target protocol.	Positive values indicate an improvement over the baseline on that target.	Not equivalent to a generalization gap; by itself, it does not establish generalization beyond training exposure.

Table 15: Evaluation metrics and protocols for reasoning generalization across training, architecture, inference, and analysis. This catalogue focuses on broadly useful measures for interpreting and comparing evidence, rather than exhaustively listing paper-specific metrics. It covers task performance, sampling and selection, verification, computational cost, and survey-defined quantities for cross-study analysis. Meaningful comparisons require sufficiently aligned evaluation protocols and resource budgets; ID/OOD status is determined by the evaluation protocol and training exposure, not by the benchmark name alone.